

TACET: Tacit, Anchored-Context, Epoch-free Threshold Encryption for Encrypted Mempools

Luis [SURNAME]

brainbot GmbH · Shutter Network

working draft

June 2026

Abstract

We give a concrete design for a threshold encryption scheme aimed squarely at the encrypted-mempool desiderata collected by Wagner. The starting observation is that no single published scheme meets the whole list at once: silent setup, non-interactive and computationally cheap decryption, CCA security with associated data, security against rogue ciphertexts, context-dependent decryption shares, epoch-free batching, small ciphertexts and shares, and a plausible post-quantum story. TACET is a synthesis that meets most of them simultaneously. Its organising idea is a single per-block *anchor* string that plays three roles at once: it is the decryption *context* that isolates partial decryptions across forks (so shares cast for a losing proposal cannot open the canonical block), it is the *selector* that defines the epoch-free batch (so only transactions that are actually included are ever revealed), and it is bound by committee members at the moment they vote, so the batch opening is ready at finalisation with no extra latency round. We pair this with an encryptor-chosen associated-data field that keeps the hybrid KEM/DEM chosen-ciphertext secure. We give two instantiations. The first is over pairings, building on the silent-setup line of Garg–Kolonelos–Policharla–Wang and BEAST-MEV, the epoch-free batching of BEAT-MEV with the FFT speed-up of the weighted variant, the context-isolation of Boneh–Bünz–Nayak–Rotem–Shoup, and the CPA-to-CCA framework of Brzuska–Kloof–Woo; it is silent but classical. The second is *fully post-quantum*, from Module-LWE, and is the main result: a thresholded Regev core with small shares and non-interactive decryption (Pilvi; Micciancio–Suhl; Boudgoust–Scholl), the anchor realised through a context-keyed *key-homomorphic* PRF together with a vector commitment that fixes the included batch so only included transactions open, and CCA2 plus robustness from straight-line simulation-extractable lattice NIZKs (Lyubashevsky–Nguyen–Plançon; LaBRADOR) in the quantum random oracle model. For this construction we give a formal security model, formal QROM theorems with structured proofs (the per-hop reductions cited, the composition and noise budget machine-checked in Coq), a sound, deployable concrete parameter set (a 7-of-13 committee) whose noise budget closes at a $\approx 2^{31}$ modulus with failure far below 2^{-128} , and a reference implementation — with a constant-time native (Rust) build, and a GPU stress of the noise budget at $2 \cdot 10^9$ samples — that validates the algorithms, the security properties, and the noise budget end to end. The post-quantum scheme makes the setup transparent and meets the post-quantum desideratum, but it trades silent aggregation for a one-time distributed key generation, which is how real keyper committees already run, its ciphertexts are larger, and (a correction to a first draft) its batch opening is $O(|B|)$ rather than constant, the honest cost of the linear lattice route. We are deliberate about what is proved and what still needs external work, and a dual-column scorecard against Wagner’s list closes the paper.

Scope statement. TACET is a construction with two instantiations; the post-quantum one is the focus. For it we give a formal security model (Definition 2), explicit algorithms (Construction 2), a sound, deployable concrete parameter set (Table 4), formal QROM theorems with structured proofs (composition machine-checked, per-hop reductions cited) in Section A, and a reference implementation with tests, benchmarks, a concrete-security estimate, a real zero-knowledge well-formedness NIZK, and a constant-time native build (Section 10). We are equally explicit about what remains external work (Section 12): a *batch-aggregated* succinct NIZK (LaBRADOR; the implemented proof is sound and zero-knowledge but linear-size), a like-for-like comparison against BEAST-mev’s optimised code, and independent or machine-checked verification of the proofs (the noise composition, the context-isolation cancellation, and the flooding hop’s order-2 probability bound are machine-checked in Coq; the concrete-security figure is confirmed by the community-standard estimator under a gate-count model). What we still mark **Open** as research is narrower: a *silent* post-quantum setup and fully adaptive corruptions for the batched lattice scheme. This is a rigorous draft with its residual gaps marked, not a finished, refereed result.

1 Desiderata

An encrypted mempool hides transactions until their inclusion is final, removing the window in which a transaction is public but not yet ordered, which is the window front-running and sandwiching exploit. The usual template is: a

user broadcasts a threshold-encrypted transaction; it is included in block i ; a committee of keypers releases short decryption material; the transaction is revealed and executed in block $i + 1$.

Wagner’s literature note distils what a good threshold scheme for this template should provide. We restate the list as desiderata, keeping his terminology, and refer to them throughout.

D1 Setup

No trusted setup; transparent setup preferred.

D2 Silent setup

No distributed key generation (DKG). Parties post local public keys; the joint key is a deterministic function of them. Important once the committee is large.

D3 Decryption

Non-interactive: one round of partial decryptions, no keyper-to-keyper messaging.

D4 CCA2

Adaptive chosen-ciphertext security, not merely CPA or CCA1.

D5 Ciphertext-adaptive

Security holds for adaptively chosen challenge ciphertexts.

D6 Corruption-adaptive

Static corruptions acceptable, adaptive preferred.

D7 Rogue ciphertexts

In the batched setting, a malformed or adversarial ciphertext must not leak anything about honest ciphertexts in the same batch.

D8 Post-quantum

A plausible post-quantum instantiation is a plus.

D9 Context-dependent shares

Decryption shares carry a context; shares produced under different contexts cannot be combined.

D10 Small public key

Small per-party public key in the silent-setup model.

D11 Small ciphertext

Low ciphertext size and overhead.

D12 Short decryption keys

A short symmetric key suffices to open a transaction, via hybrid encryption; this needs a threshold KEM *with associated data*, otherwise CCA2 is not preserved. Threshold IBE is one route.

D13 Small shares

Small decryption shares.

D14 Batch decryption

Release one short key for many ciphertexts. Epoch-free (no per-block setup and no slot number) is strongly preferred over epoch-based.

D15 Fast decryption

Decryption is computationally cheap; it sits on the critical path of block production.

Contribution. We present TACET, a design that satisfies D2, D3, D4, D5, D7, D9, D11, D12, D13, D14, D15 and partially satisfies D1, D6, D10 in its pairing instantiation, and that, in a second *post-quantum* instantiation from Module-LWE, additionally satisfies D8 and makes D1 fully transparent, at the cost of D2 (now a one-time DKG), a larger ciphertext (D11) and an $O(|B|)$ batch opening (D13 becomes partial). The new ingredients are: (i) a single anchor that unifies the decryption context (D9), the epoch-free batch selector (D14), and the consensus binding that removes the extra decryption round; (ii) two ways to make the anchor isolate shares, one per instantiation, namely the observation that in silent-setup pairing schemes a partial decryption is a signature share so anchoring its message gives context isolation for free, and a lattice analogue where a context-keyed pseudorandom *zero-sharing* gives the same isolation because its masks cancel within a context but leave an unremovable pseudorandom residue when shares from different contexts are combined; (iii) a de-risked post-quantum construction (Section 8 and Construction 2) that avoids the fragile key-homomorphic *puncturable* PRF a naive port would need, getting the only-included property (D14) from committee scoping plus a vector commitment that fixes the batch, isolating

contexts with a pseudorandom *zero-sharing* whose masks cancel under the low-norm recovery coefficients (so the masking adds no noise) while the core secret’s low-norm sharing keeps a single noise budget in control; (iv) a formal security model (Definition 2), formal QROM theorems (Theorems 4 to 6) with proofs (Section A), and a derived concrete parameter set (Table 4) whose noise budget and simulation requirement close together; and (v) a careful, sourced, dual-column scorecard that states honestly where each instantiation stands and what remains external work.

2 Background and landscape

We sketch the four lines TACET draws on, then position the published schemes against the desiderata in Table 1.

Silent setup. Garg, Kolonelos, Policharla and Wang (GKPW, CRYPTO 2024) build threshold encryption where the joint public key is a deterministic function of locally generated per-party keys, so the setup is silent: asynchronous, with multiverse support and a dynamic threshold. The core is a special-purpose witness encryption for the statement “at least t of the registered parties signed a given message,” over a Kate–Zaverucha–Goldberg (KZG) structured reference string, proved secure in the generic group model. This is the silent ancestor of the rest.

Epoch-free batching. BEAT-MEV (Bormet–Faust–Othman–Qu, USENIX Security 2025) makes batching epoch-free: one one-time setup supports all future batches, and crucially only the transactions actually included in a block are revealed, closing the “MEVade” leak of Choudhuri–Garg–Piet–Policharla in which epoch-based schemes reveal transactions that were submitted but never included. The mechanism is a key-homomorphic puncturable PRF (KH-PPRF) over an additively homomorphic threshold encryption. BEAT-MEV (Bormet–Choudhuri–Faust–Garg–Othman–Policharla–Qu–Wang) adds a *silent* setup to epoch-free batching by making the GKPW scheme additively homomorphic; it is CCA secure under static corruptions via a straight-line simulation-extractable NIZK in the algebraic group model, and it is concretely efficient with a public Rust implementation. The weighted variant (Agarwal et al.) replaces the KH-PPRF with an FFT-friendly one and lowers batch decryption from $O(B^2)$ to $O(B \log B)$ in the batch size B .

Context-dependent shares. Boneh, Bünz, Nayak, Rotem and Shoup (BBNRS, ASIACRYPT 2025) introduce a *decryption context*: an extra argument to partial decryption such that shares produced under different contexts cannot be combined. Concretely, if fewer than t shares are produced under context A and fewer than t under context B , the ciphertext stays hidden even if the total exceeds t . Their Diffie–Hellman-group constructions are non-interactive and withstand adaptive corruptions, and the headline application is exactly ours: keyers attach decryption shares to their consensus vote on a block, which minimises latency, while the context keeps forks isolated.

CCA, associated data and the succinctness barrier. Brzuska, Klooß and Woo (BKW) give the first systematic study of threshold-PKE confidentiality: IND versus SIM, CPA versus CCA, static versus adaptive, plus two generic CPA-to-CCA transforms. One insight we use directly: security under *maximal* corruptions ($t - 1$) does not imply security under fewer corruptions once the adversary can see partial decryptions of the challenge, which is exactly the regime an encrypted mempool runs in. This is what makes a context-dependent (“high-threshold”) notion the right target rather than a convenience. Separately, Boneh, Laufer and Tas (BLT) prove a *succinctness vulnerability*: if the per-batch decryption key for a set is made too short, security breaks; their positive results give lattice and KP-ABE batch IBE. We respect this barrier in Section 5.

Post-quantum threshold decryption. The lattice line splits into two routes with very different decryption profiles, and conflating them is the mistake that makes post-quantum threshold encryption look harder than it is. The *trapdoor* route thresholdises a Gaussian preimage sampler, as in the lattice batch IBE of BLT; sampling a preimage from key shares is interactive, so decryption is two-round, which conflicts with D3. The *linear* route goes back to Bendlin–Damgård: secret-share the Regev key, let each keyper publish a noisy inner product, and add and round. This is *non-interactive* (D3), and the only cost is noise flooding to hide the key. Recent work tames that cost: Boudgoust–Scholl and Micciancio–Suhl bring the modulus down to polynomial (Rényi and Gaussian-flooding analyses) with parameters comparable to FrodoKEM, and Pilvi (Cini–Lai–Woo, ASIACRYPT 2025) gives decryption shares of size $t \cdot \log K \cdot \text{poly}(\lambda)$ that stay secure *even when the adversary sees partial decryptions of the challenge*, which is exactly the BKW regime an encrypted mempool runs in. For the proof layer, lattice Fiat–Shamir NIZKs (Lyubashevsky–Nguyen–Plançon, CRYPTO 2022; LaBRADOR for succinctness) now come with straight-line, simulation-extractable variants in the random oracle model, which is what a post-quantum

Scheme	Silent	No tr. setup	Epoch-free	CCA	Ctx shares	Adaptive corr.	PQ	Fast dec.
GKPW '24 (2024/263)	Yes	No	n/a	Partial	No	No	No	Yes
BEAT-MEV (2024/1533)	No	No	Yes	Yes	No	No	No	Partial
BEAST-MEV (2025/1419)	Yes	No	Yes	Yes	No	No	No	Yes
Weighted (2025/2115)	No	No	Yes	Yes	No	No	No	Yes
BBNRS (2025/279)	No	Yes	n/a	Yes	Yes	Yes	No	Yes
BLT lattice (2025/1254)	No	Partial	Yes	Partial	No	No	Yes	Partial
Champion–Wu (2026/318)	Yes	Yes	n/a	Partial	No	No	Yes	Partial
GongWWW (2025/2103)	Yes	No	Partial	Partial	No	No	No	Yes
Pilvi (2025/1691)	No	Yes	n/a	Partial	No	No	Yes	Partial
TACET-P	Yes	Partial	Yes	Yes	Yes	Partial	No	Yes
TACET-Q	No	Yes	Yes	Yes	Yes	Partial	Yes	Partial

Table 1: Published schemes against the headline desiderata, with the two TACET instantiations for comparison. Characterisations follow each paper and Wagner’s note; “n/a” marks unbatched schemes where epoch-freeness is not defined. **Partial** for CCA on GKPW/Champion–Wu/GongWWW reflects tag-based or CPA-leaning notions; **Partial** on “no trusted setup” marks an updatable powers-of-tau SRS rather than a fully transparent one. TACET-P is the pairing instantiation, TACET-Q the post-quantum one; the lattice column flips for TACET-Q at the cost of the silent column.

CCA lift needs. What the lattice line still lacks is *silent* setup: the only silent lattice object is distributed broadcast encryption (the $t = 1$ case), and only selectively secure. General silent t -of- n threshold decryption from a falsifiable lattice assumption is open; TACET-Q therefore uses a one-time DKG, which matches deployed keyper committees.

The gap is visible in the last two columns of Table 1. The pairing line (BEAST-MEV) gives silent, epoch-free, CCA, fast decryption, but is static and not post-quantum. The DDH line (BBNRS) gives context isolation and adaptive corruptions, but is unbatched. The lattice line is the subtle one: the trapdoor route (BLT) is post-quantum and transparent but needs interactive decryption, whereas the linear Regev route (Pilvi) is post-quantum, transparent, *and* non-interactive, and what it lacks is batching, context and silence. TACET-P combines the first three lines; TACET-Q adds batching, context and CCA to the Pilvi route and so closes the post-quantum gap, paying for it with a one-time DKG in place of silent setup.

3 Definitions and security model

We work with a threshold KEM rather than a full encryption scheme, because D12 asks for a short symmetric key released per transaction, and BKW shows a threshold KEM *with associated data* is what keeps the hybrid CCA2. The DEM is any standard AEAD. The model below is the object both instantiations realise; our formal results (Sections A and 8) are stated for the post-quantum construction TACET-Q, which is the main contribution, with the pairing construction TACET-P kept as a silent classical companion.

Definition 1 (Anchored batched threshold KEM with associated data). For n keypers and threshold t , an *anchored batched threshold KEM with associated data* is a tuple of algorithms:

- $\text{Setup}(1^\lambda) \rightarrow \text{crs}$: outputs a public reference string.
- $\text{KGen}(\text{crs}, i) \rightarrow (\text{sk}_i, \text{pk}_i)$: party i samples a secret key and derives its public key locally, with no interaction.
- $\text{AggKey}(\text{crs}, \{\text{pk}_i\}_{i \in [n]}, t) \rightarrow (\text{ek}, \{\text{vk}_i\}_{i \in [n]})$: deterministically aggregates the public keys into a short encapsulation key ek and per-party verification keys.
- $\text{Encap}(\text{ek}, \text{ad}) \rightarrow (\text{ct}, K)$: outputs a ciphertext header ct and a key $K \in \{0, 1\}^\lambda$, binding the encryptor-chosen associated data ad .
- $\text{Verify}(\text{ek}, \text{ct}, \text{ad}) \rightarrow \{0, 1\}$: public well-formedness check on a ciphertext.
- $\text{PartDec}(\text{sk}_i, \text{ctx}, B) \rightarrow \rho_i$: party i , given an anchor ctx and a batch $B = \{\text{ct}_1, \dots, \text{ct}_{|B|}\}$ of well-formed headers, outputs a partial opening ρ_i bound to ctx , of size independent of $|B|$.
- $\text{Open}(\text{ek}, \text{ctx}, B, \{\rho_i\}_{i \in S}) \rightarrow \Delta$: from any t valid partial openings under the *same* ctx , outputs a single batch opening Δ .
- $\text{Decap}(\text{ek}, \text{ctx}, B, \Delta, \text{ct}_j) \rightarrow K_j$: recovers the key of each $\text{ct}_j \in B$.

The anchor. ctx is a digest of the proposed block at a given height, for instance its header or the Merkle root of the encrypted transactions it includes. It is chosen by the committee at decryption time, not by the encryptor. Two distinct mechanisms bind to two distinct fields, and conflating them is a common mistake, so we are explicit:

- ad is *encryptor-chosen* at encryption time and is bound into ct and the AEAD. It is what makes the hybrid CCA2 (D12). The encryptor cannot bind a block: it does not yet know which block will include the transaction.
- ctx is *committee-chosen* at decryption time. Partial openings are bound to it. It isolates shares across forks (D9), and it selects the batch (D14).

Correctness. For honest keys, any well-formed $\text{ct}_j \in B$, and any t -subset S of honest partial openings under ctx , Decap returns the K_j output by Encap, except with probability at most $\delta_{\text{corr}}(\lambda)$ (the decryption-failure rate, bounded concretely in Section 8.4).

Confidentiality. We use a static-corruption, simulation-flavoured indistinguishability game with a per-context threshold restriction (we call it IND-CCA-CTX). It is the game an encrypted mempool actually runs in: keypers publish partial openings, the adversary harvests them across forks, and it may push malformed ciphertexts and ask for decryptions.

Definition 2 (IND-CCA-CTX). For a scheme Π and adversary $\mathcal{A}$, the experiment $\text{Exp}_{\Pi, \mathcal{A}}^{\text{IND-CCA-CTX}}(\lambda)$ is:

1. $\text{crs} \leftarrow \text{Setup}(1^\lambda)$; $\mathcal{A}$ outputs a corrupt set $C \subseteq [n]$, $|C| \leq t - 1$.
2. For $i \notin C$, $(\text{sk}_i, \text{pk}_i) \leftarrow \text{KGen}(\text{crs}, i)$; for $i \in C$, $\mathcal{A}$ supplies pk_i . Compute $(\text{ek}, \{\text{vk}_i\}) \leftarrow \text{AggKey}(\text{crs}, \{\text{pk}_i\}, t)$ and give $\text{ek}, \{\text{vk}_i\}, \{\text{sk}_i\}_{i \in C}$ to $\mathcal{A}$.
3. $\mathcal{A}$ gets oracles, with a query log:
 - $\mathcal{O}_{\text{PartDec}}(i, \text{ctx}, B)$ for honest i : return $\text{PartDec}(\text{sk}_i, \text{ctx}, B)$ and record (ctx, B, i) .
 - $\mathcal{O}_{\text{Decap}}(\text{ctx}, B, \{\rho_j\})$: return Open then Decap on valid shares.
4. $\mathcal{A}$ submits ad^* ; challenger draws $(\text{ct}^*, K^*) \leftarrow \text{Encap}(\text{ek}, \text{ad}^*)$, a bit b , sets $K_0 := K^*$, $K_1 \xleftarrow{\$} \{0, 1\}^\lambda$, returns (ct^*, K_b) .
5. $\mathcal{A}$ keeps oracle access and outputs b' .

Admissibility. For every ctx such that $\mathcal{A}$ obtains an $\mathcal{O}_{\text{PartDec}}(\cdot, \text{ctx}, B)$ opening with $\text{ct}^* \in B$, the number of distinct honest indices queried under that ctx plus $|C|$ is at most $t - 1$; and $\mathcal{A}$ never Decap-queries a batch containing ct^* under such a ctx .

The advantage is $\text{Adv}_{\Pi, \mathcal{A}}^{\text{IND-CCA-CTX}}(\lambda) = \left| \Pr[b' = b] - \frac{1}{2} \right|$, and Π is IND-CCA-CTX secure if it is $\text{negl}(\lambda)$ for all probabilistic polynomial-time $\mathcal{A}$ making at most Q decryption queries.

This one game carries four of Wagner’s requirements at once: the Decap oracle gives CCA (D4); ad^* is chosen adaptively (D5); the *per-context* counting in the admissibility clause is exactly context-dependence (D9); and allowing up to $t - 1$ challenge-batch openings per context is the see-the-challenge-shares regime that BKW single out. The quantum version $\text{Adv}_{\Pi, \mathcal{A}}^{\text{IND-CCA-CTX}}$ gives $\mathcal{A}$ quantum access to any random oracle; this is the notion Theorem 4 targets, and the distinction matters (a classical-ROM proof does not certify post-quantum security).

Definition 3 (Context isolation). Π has *context isolation* if there is an efficient simulator that, for any $\text{ctx}' \neq \text{ctx}$, produces openings under ctx' from public data and $\{\text{sk}_i\}_{i \in C}$ alone, with a distribution indistinguishable from honest $\mathcal{O}_{\text{PartDec}}(\cdot, \text{ctx}', \cdot)$ answers, in the view of an adversary attacking a batch under ctx . Hence openings under ctx' never count toward the threshold under ctx : fewer than t openings under each of ctx, ctx' cannot open a ciphertext even if their union exceeds t .

This is what makes the per-context clause in Definition 2 the *only* restriction, and it is the formal content of Remark 1: an adversary that withholds the canonical block’s opening but harvests sub-threshold openings from competing proposals gains nothing.

Definition 4 (Rogue-ciphertext security and robustness). (*Rogue ciphertexts, D7.*) For any batch B mixing honest ciphertexts with adversarial $\text{ct}^\dagger$, the opening Δ for B leaks nothing about the honest ciphertexts beyond their own keys K_j ; formally, Definition 2 still holds when $\mathcal{A}$ chooses the non-challenge members of every queried batch. (*Robustness / identifiable abort.*) If a published share ρ_i fails its verification relation, Open rejects it and outputs the index i ; given any t valid shares under ctx , Open returns the correct Δ .

Remark 1 (Why anchored contexts, not just maximal-corruption CCA). BKW show that proving security only at the maximal corruption level $t - 1$ is insufficient once partial decryptions of the challenge are available, which is precisely the mempool setting where keypers publish openings. Definition 3 is the strengthening that closes this gap, and Definition 2 bakes it into the admissibility clause rather than assuming maximal corruption.

4 The TACET construction

4.1 One anchor, three jobs

The design rests on a single observation. In silent-setup schemes of the GKPW family, a ciphertext is encrypted to the witness-encryption statement “at least t keypers signed message μ ,” and a keyper’s partial decryption is a signature share on μ . Signature shares on two different messages do not combine into a valid signature on either. Therefore, if we let the batch-opening statement be anchored to ctx , partial openings for ctx and for $\text{ctx}' \neq \text{ctx}$ are non-combinable by construction. Definition 3 then comes for free from the unforgeability that already underlies the silent-setup scheme, with no extra context machinery of the BBNRS kind. The same ctx names the batch, so it is also the epoch-free selector, and committee members can sign it as part of their vote. One string, three jobs.

The same three jobs survive the move to lattices, by a different mechanism. There a partial decryption is not a signature share but a noisy linear value, so we bind ctx through the batching layer instead: the per-ciphertext mask is produced by a key-homomorphic puncturable PRF evaluated at ctx , so shares cast under ctx and ctx' evaluate the PRF at different inputs and their masks do not cancel when combined. Isolation then follows from PRF pseudorandomness rather than signature unforgeability. We develop this in Section 8.

4.2 Layers

TACET is built in four layers; each cites the primitive it instantiates.

Layer A: silent-setup homomorphic threshold KEM. Take the GKPW silent-setup threshold encryption and use it as a KEM: AggKey is GKPW’s deterministic aggregation, ek is the joint key, and K is derived from the witness-encryption payload. Following BEAST-MEV, make the scheme additively homomorphic in the $\mathbb{G}_2$ component, which is what the batching layer needs. Static-corruption CPA holds in the generic group model for GKPW and is preserved by the homomorphic variant.

Layer B: epoch-free batching. Attach to each ciphertext a per-ciphertext index and a KH-PPRF-derived mask, as in BEAT-MEV. At decryption the committee produces a single opening for the set B present under ctx ; the homomorphism plus the public set B let any decryptor compute every K_j for $j \in B$, and the puncturing prevents opening any index outside B . Using the FFT-friendly KH-PPRF of the weighted variant, the decryptor’s work is $O(|B| \log |B|)$ rather than $O(|B|^2)$. The opening Δ is constant size in $|B|$, so committee-to-chain communication is constant (D13, D14, D15).

Layer C: anchored contexts. Set the batch-opening statement to bind ctx , so that $\text{PartDec}(\text{sk}_i, \text{ctx}, B)$ is keyper i ’s signature share on (a digest of) (ctx, B) . By §4.1 this yields Definition 3. Operationally, keyper i emits ρ_i inside its consensus vote for the block whose header is ctx ; once the block is finalised, t such votes already carry t valid ρ_i , so Δ is available with no extra round. A share emitted for a competing proposal carries a different ctx and is inert against the canonical block. This is the BBNRS latency story, realised through the signature-share structure already present in Layer A.

Layer D: CCA2 with associated data, and robustness. Lift the CPA core to CCA2 with the BKW machinery. Concretely, Encap attaches a straight-line simulation-extractable NIZK that ct is well-formed with respect to ad ; the statement includes ad , and the AEAD binds ad , so the hybrid is CCA2 and cut-and-paste of a header into a different ad is rejected (D4, D5, D12). The same proof rejects malformed ciphertexts, giving Definition 4 (D7). Each ρ_i carries a share-verifiability NIZK so that Open uses only valid shares; this gives output robustness and blocks denial of service by malformed shares, as in BEAST-MEV.

4.3 Concrete instantiation and parameters

- **Groups.** A type-3 pairing $e : \mathbb{G}_1 \times \mathbb{G}_2 \rightarrow \mathbb{G}_T$, for example BLS12-381.
- **Setup.** A universal, updatable powers-of-tau (KZG) SRS, size $O(n)$ in the committee. This is reusable and ceremony-based, not per-scheme, but it is *not* transparent (D1 is only partially met; see Section 7).
- **Idealisations.** Generic group model for the Layer A core, algebraic group model and random oracle for the batching and CCA NIZKs. This matches the published analyses of the components and is stated plainly rather than hidden.

- **Assumptions.** The q -type pairing assumption underlying the KZG-based witness encryption of Layer A; pseudorandomness of the KH-PPRF of Layer B; a DDH-type assumption for the additive masking; and knowledge-soundness of the NIZKs of Layer D.

Construction 1 (TACET-P, pairing instantiation). Instantiate Definition 1 with Layer A = additively homomorphic GKPW, Layer B = FFT-friendly KH-PPRF batching, Layer C = ctx-anchored signature-share openings, Layer D = associated-data NIZK lift plus share verifiability, over a type-3 pairing with an updatable KZG SRS. The post-quantum instantiation TACET-Q is Construction 2 in Section 8.

5 Security

We argue security modularly. Each statement reduces to the published result for the relevant layer; we are explicit that the end-to-end proof for the composed scheme is future work (§12).

Theorem 1 (Confidentiality, informal). *Under Construction 1, no probabilistic polynomial-time adversary that statically corrupts at most $t - 1$ keypers and controls fewer than t partial openings in every context containing the challenge wins the SB-IND-CCA game with non-negligible advantage, assuming: CPA security of the Layer A homomorphic core (generic group model), pseudorandomness of the Layer B KH-PPRF, and straight-line simulation-extractability of the Layer D NIZK (algebraic group and random oracle models).*

Argument. The reduction follows the BEAST-MEV proof: simulate the Layer D NIZKs, extract from any decryption-oracle query to answer Decap and Open without the secret keys, and reduce a distinguisher on K^* to a CPA distinguisher on the homomorphic core, where the KH-PPRF masks of non-challenge ciphertexts are replaced by random by pseudorandomness. The corruption bound is static and inherited from Layer A. $\square$

Theorem 2 (Context isolation). *Under Construction 1, Definition 3 holds, assuming unforgeability of the threshold signature underlying Layer A.*

Argument. A successful cross-context combination yields t valid signature shares on a single anchored message from fewer than t shares on each of two distinct messages, contradicting unforgeability. Note this property is about which block's batch can be opened; it does not require the encryptor to bind a block, and indeed it cannot (Layer B is epoch-free). $\square$

Theorem 3 (Rogue-ciphertext security). *Under Construction 1, Definition 4 holds, assuming knowledge-soundness of the Layer D well-formedness NIZK and per-index puncturing of the KH-PPRF.*

Remark 2 (Respecting the succinctness barrier). BLT show that a per-batch *set* key that is too short is insecure. TACET-P sidesteps this: Δ is constant size as committee-to-chain *communication*, but the per-ciphertext overhead stays constant and each member's ct_j carries its own masking term, so the set is not encoded into a sub-barrier object. The lattice route TACET-Q does *not* share the pairing line's succinct opening: its opening is $O(|B|)$ ring elements, one per included ciphertext (Construction 2); a single $\mathcal{R}_q$ element cannot encode the $|B|$ independent recovered values, and there is no lattice analogue of the pairing/witness-encryption structure that makes the pairing opening constant. We make no claim of sublinear succinctness for set membership in either instantiation, and any future move toward more succinct openings must be re-checked against BLT.

Remark 3 (Adaptive corruptions). Layer A is static, so Theorem 1 is static. The Layer C/D context and CCA structure is compatible with the adaptive, non-interactive techniques of BBNRS in the unbatched case. A fully adaptive *batched* silent-setup scheme remains **Open**: it requires an adaptively secure homomorphic core in the silent-setup model, which is not currently known.

6 Efficiency and the mempool flow

Table 2 summarises asymptotic costs; constants follow the BEAST-MEV and weighted-variant implementations. The salient points for D11, D13, D14, D15 are: a ciphertext is a constant number of group elements plus a short proof; a partial opening is constant and independent of batch size; the batch opening is constant in B ; and batch decapsulation is $O(B \log B)$.

Zero extra latency. Figure 1 shows the flow. Because keypers attach ρ_i to their vote on block i under anchor ctx_i , the moment block i finalises with t supporting votes, t valid openings exist; Open and Decap run and the batch is revealed for execution in block $i + 1$ without a dedicated decryption round. Forks are handled by Definition 3: a vote for a competing proposal binds a different ctx and cannot help open block i , and conversely sub-threshold openings of block i do not leak if block i never finalises.

Object / step	Cost
SRS (crs)	$O(n)$ group elements, universal and updatable
Per-party public key pk_i	$O(1)$ group elements
Aggregation / verification material $\{vk_i\}$	$O(n)$ total (the “large keys” caveat of this line)
Encapsulation key ek	$O(1)$ group elements
Ciphertext header ct	$O(1)$ group elements + short well-formedness proof
Symmetric key K	λ bits (hybrid via AEAD)
Partial opening ρ_i	$O(1)$, independent of $ B $, + share-verifiability proof
Batch opening Δ	$O(1)$ in $ B $ (constant committee-to-chain communication)
PartDec (per keyper)	$O(B)$ field operations
Decap (per block, public)	$O(B \log B)$ with the FFT-friendly KH-PPRF

Table 2: Asymptotic costs of TACET in the pairing instantiation. n is the committee size, B the batch, λ the security parameter.

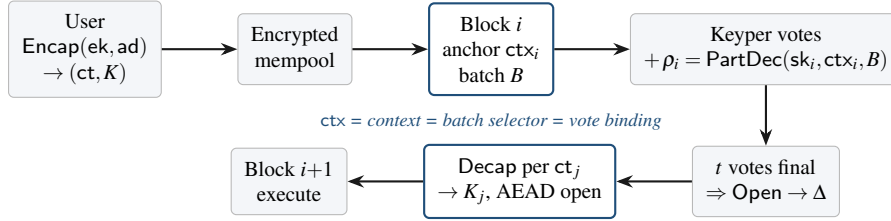


Figure 1: The encrypted-mempool flow. The single anchor ctx_i is the decryption context, the epoch-free batch selector, and the binding under which keypers vote, so decryption material is ready at finalisation.

7 Property scorecard

Table 3 maps each desideratum to the status of both instantiations with a one-line reason. This is the honest centre of the paper: every **Partial**, **No**, and **Open** is stated, not glossed. The pairing column TACET-P meets 11 of 15; the post-quantum column TACET-Q meets 10 fully and D2 partially (its decryption-share size D13 is **Partial**, because the batch opening is $O(|B|)$ on the linear lattice route — the correction of Remark 2). TACET-Q moves D8 from **No** to **Yes** and makes D1 fully transparent; as *built* it uses a one-time DKG for setup (D2), but a silent setup is now achievable post-quantum from a falsifiable assumption and prototyped in `silentpq/` (Section 8.6), so D2 is **Partial** rather than **No**. The residual costs are a concretely larger ciphertext (D11) and the $O(|B|)$ opening (D13).

8 The post-quantum construction TACET-Q

This is the main contribution. We give a concrete post-quantum instantiation of Definition 1 from Module-LWE, with explicit algorithms, a closed noise budget at concrete parameters (Section 8.4), and formal security in the quantum random oracle model (Section A). The structure follows BEAST-MEV with lattice substitutions: an additively homomorphic threshold-Regev core in place of the pairing core, a context-keyed pseudorandom zero-sharing in place of the pairing KH-PPRF, and a one-time DKG in place of silent setup. A reference implementation (Section 10) validates the algorithms, the security properties, and the noise budget end to end.

Three de-risking changes. A naive port would ask for a key-homomorphic *puncturable* PRF over lattices, to get the “only included transactions are revealed” property (D14) by puncturing at the excluded indices. We avoid that object, because a lattice PRF that is at once key-homomorphic, puncturable, low-error and ring-efficient is not known at usable parameters, and the puncturing interacts badly with the flooding budget. (i) Instead the anchor ctx *commits* to the included batch through a vector commitment, and “only-included” follows from the committee scoping its partial openings to $j \in B$: an excluded ciphertext receives no opening, so its key is never produced. The vector commitment’s job is to *publicly fix* the set B (so a decapsulator cannot substitute a different set or position), not to gate the key cryptographically; no puncturing is needed. (ii) The batch opening is $O(|B|)$, one $\mathcal{R}_q$ element per included ciphertext: a single $\mathcal{R}_q$ element cannot encode the $|B|$ independent values $\mathbf{s}^\top \mathbf{u}_j$ a decapsulator must recover, so the constant-size opening of the pairing line (BEAST-MEV) has no lattice analogue. This respects the succinctness barrier (Remark 2) and is the honest cost of the linear route. (iii) Context isolation is realised by a context-keyed pseudorandom *zero-sharing* whose masks cancel under the recovery coefficients but

	Desideratum	-P	-Q	Reason (P pairing, Q post-quantum)
D1	No trusted setup	Partial	Yes	P: updatable powers-of-tau SRS. Q: uniform matrix from a public seed, transparent.
D2	Silent setup	Yes	Partial	P: GKPW deterministic aggregation. Q: one-time dealerless DKG <i>as built</i> , but a silent setup is now achievable from falsifiable ℓ -succinct/decomposed LWE (Champion–Wu TCC24/EC26; prototyped in <code>silentpq/</code> , Section 8.6) and adoptable to drop the DKG.
D3	Non-interactive dec.	Yes	Yes	One round of partial openings; Q adds and rounds (Pilvi), no keyper-to-keyper messaging.
D4	CCA2	Yes	Yes	Associated-data NIZK lift; AGM proof for P, QROM lattice NIZK for Q.
D5	Ciphertext-adaptive	Yes	Yes	Decryption oracle via straight-line extraction; Q within an a priori query bound Q .
D6	Corruption-adaptive	Partial	Partial	Static for both batched cores; fully adaptive batched is Open (Remark 3).
D7	Rogue ciphertexts	Yes	Yes	Public well-formedness NIZK plus per-ciphertext membership gating (Theorem 6).
D8	Post-quantum	No	Yes	P is pairing-based; Q is Module-LWE confidentiality, Module-SIS soundness.
D9	Context-dependent shares	Yes	Yes	P: signature-share anchoring. Q: anchor-keyed KH-PRF; shares under different anchors do not combine.
D10	Small public key	Partial	Partial	$O(1)$ encapsulation key; $O(n)$ aggregation (P) or DKG (Q) material.
D11	Small ciphertext	Yes	Partial	P: $O(1)$ group elements. Q: a Regev pair plus a short proof, constant but FrodoKEM-scale, the PQ price.
D12	Short decryption keys	Yes	Yes	λ -bit K via AEAD; associated data bound, the object BkW prescribes.
D13	Small shares	Yes	Partial	P: constant in $ B $. Q: per-ciphertext opening is one $\mathcal{R}_q$ element (≈ 1 KB); the batch opening is $O(B)$, not constant (no lattice analogue of the pairing-line succinct opening; Remark 2).
D14	Epoch-free batching	Yes	Yes	BEAT/BEAST (P); Q binds the batch with a vector commitment and the committee scopes openings to B , so only included transactions are revealed (no puncturing needed).
D15	Fast decryption	Yes	Yes	P: $O(B \log B)$. Q: per-ciphertext Decap is a subtraction and rounding (measured 0.14 ms, Section 10); proofs verify at mempool admission, off the critical path.

Table 3: The two TACET instantiations against Wagner’s desiderata: TACET-P meets eleven, TACET-Q ten (D13 **Partial**: $O(|B|)$ batch opening). The post-quantum column trades the silent setup (D2), a larger ciphertext (D11) and the $O(|B|)$ opening (D13) for post-quantum security (D8) and a transparent setup (D1). Remaining **Open** items are in Section 8.6 and remark 3.

not across contexts (Section 8.3); this is what makes the masked openings both decrypt correctly and isolate forks, and it makes the key-homomorphism error vanish in the exact realisation. All three are validated by the reference implementation of Section 10.

8.1 Building blocks

- **Ring and core.** $\mathcal{R} = \mathbb{Z}[X]/(X^N + 1)$, $\mathcal{R}_q = \mathcal{R}/q\mathcal{R}$ with N a power of two. The KEM core is Lindner–Peikert over a module of rank k : public $\mathbf{A} \in \mathcal{R}_q^{k \times k}$ expanded from a seed, $\mathbf{b} = \mathbf{A}\mathbf{s} + \mathbf{e}$ with short $\mathbf{s}, \mathbf{e}$. The threshold secret is $\mathbf{s}$. Encapsulation encodes a random seed $\mu \in \{0, 1\}^N$ as $\lfloor q/2 \rfloor \cdot \mu$, producing $\mathbf{u} = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1$ and $w = \mathbf{b}^\top \mathbf{r} + e_2 + \lfloor q/2 \rfloor \mu$; the key is $K = G(\mu, \text{ad})$ and decryption is the linear $w - \mathbf{s}^\top \mathbf{u}$ followed by coefficientwise rounding to recover μ (correct while the noise stays below $q/4$, the threshold of Lemma 1). LWE makes this additively homomorphic, which the batching layer uses.
- **Low-norm threshold sharing (Pilvi).** $\mathbf{s}$ is shared by the linear secret sharing of PILVI, whose recovery coefficients $\lambda_{i,S}$ have small norm γ and whose shares have size $t \cdot \log K \cdot \text{poly}(\lambda)$ (D13). Small recovery norm is what keeps the summed error controlled (Lemma 1); a generic Shamir sharing over $\mathbb{Z}_q$ has recovery coefficients of size $\Theta(q)$ and blows the budget, which we confirm empirically (Section 10).
- **Flooding.** Each partial decryption adds a discrete Gaussian Flood of width σ_{fl} . By the Rényi analyses of Boudgoust–Scholl and Micciancio–Suhl, a polynomial $\sigma_{\text{fl}}/\sigma_{\text{dec}}$ makes honest partial decryptions simulatable under the see-the-challenge-shares regime, for a bounded number Q of queries.
- **Context-keyed pseudorandom zero-sharing.** The per-share mask $z_i(\text{ctx}, j) \in \mathcal{R}_q$ is pseudorandom and, over the participating committee S , forms a sharing of zero against the recovery coefficients: $\sum_{i \in S} \lambda_{i,S} z_i(\text{ctx}, j) = 0$. From per-pair keys $\kappa_{i,i'}$ we set $z_i(\text{ctx}, j) = \lambda_{i,S}^{-1} \sum_{i' \in S} \text{sgn}(i, i') \text{PRF}(\kappa_{i,i'}, \text{ctx}, j)$, which cancels *exactly* (key-homomorphism error $\beta_F = 0$) by antisymmetry, while masks under distinct ctx do not cancel when mixed — the lattice analogue of BBNRS’s signature-share isolation. This replaces the almost-key-homomorphic PRF a naive port would use; an almost-KH-PRF realisation is possible but reintroduces a bounded β_F (Remark 4).
- **Vector commitment.** VC is a position-binding vector commitment (a Merkle tree, or a lattice/SIS commitment for a fully post-quantum stack). ctx includes $\text{VC}(B)$, and wit_j is the opening of position j to ct_j .
- **Lattice NIZK.** NIZK is a Fiat–Shamir lattice proof system (Lyubashevsky–Nguyen–Plançon; LaBRADOR when succinctness matters), used in its straight-line, simulation-extractable form via the Katsumata transform.

Its soundness rests on Module-SIS and it is analysed in the QROM.

8.2 The construction

Construction 2 (TACET-Q).

- $\text{Setup}(1^\lambda)$: publish a seed for $\mathbf{A} \in \mathcal{R}_q^{k \times k}$ (expanded by a random oracle), the commitment public parameters, and the zero-sharing setup. Transparent: no structured reference string.
- $\text{KGen}, \text{AggKey}$: a one-time DKG (Section 9) gives each keyper i a low-norm share $\mathbf{s}_i$ of the core secret $\mathbf{s}$ under the Pilvi sharing and the per-pair masking keys $\{\kappa_{i,i'}\}_{i'}$, and publishes $\text{ek} = (\mathbf{A}, \mathbf{b})$ and verification keys vk_i . No party learns $\mathbf{s}$.
- $\text{Encap}(\text{ek}, \text{ad})$: sample short $\mathbf{r}, \mathbf{e}_1, \mathbf{e}_2$ and a random seed $\mu \in \{0, 1\}^N$; set $\mathbf{u} = \mathbf{A}^\top \mathbf{r} + \mathbf{e}_1$ and $w = \mathbf{b}^\top \mathbf{r} + \mathbf{e}_2 + \lfloor q/2 \rfloor \mu$; the DEM key is $K = G(\mu, \text{ad})$; output $\text{ct} = (\mathbf{u}, w, \pi)$, where $\pi \leftarrow \text{NIZK}$ proves ct is a well-formed Lindner–Peikert ciphertext with short $\mathbf{r}, \mathbf{e}_1, \mathbf{e}_2$ and binds ad into the statement. Verify checks π .
- $\text{PartDec}(\mathbf{s}_i, \text{ctx}, B, S)$: with $\text{ctx} \supseteq \text{VC}(B)$ and participating committee S , output for every $j \in B$

$$\rho_{i,j} = \mathbf{s}_i^\top \mathbf{u}_j + z_i(\text{ctx}, j) + \text{Flood}_{i,j},$$

where $z_i(\text{ctx}, j)$ is the context zero-mask ($\sum_{i \in S} \lambda_{i,S} z_i(\text{ctx}, j) = 0$) and $\text{Flood}_{i,j}$ is fresh flooding. The opening $(\rho_{i,j})_{j \in B}$ is one $\mathcal{R}_q$ element per ciphertext, hence $O(|B|)$, plus a share-verifiability proof π_i bound to (ctx, B) .

- $\text{Open}(\text{ek}, \text{ctx}, B, \{\rho_{i,\cdot}\}_{i \in S})$: discard shares whose π_i fails (identifiable abort); for each $j \in B$ output $\Delta_j = \sum_{i \in S} \lambda_{i,S} \rho_{i,j}$. The masks cancel and the Pilvi coefficients reconstruct $\mathbf{s}$, so $\Delta_j = \mathbf{s}^\top \mathbf{u}_j + (\text{bounded noise})$ (Lemma 1).
- $\text{Decap}(\text{ek}, \text{ctx}, B, \{\Delta_j\}, \text{ct}_j, \text{wit}_j)$: check wit_j opens $\text{VC}(B)$ at j to ct_j ; recover μ_j by coefficientwise rounding of $w_j - \Delta_j$ and return $K_j = G(\mu_j, \text{ad}_j)$. An excluded ciphertext is not in $\text{VC}(B)$ and receives no opening, so its key is never produced (D14).

The anchor still does three jobs: it is the decryption context (it keys the zero-mask $z_i(\text{ctx}, \cdot)$, giving isolation), the batch selector (it is $\text{VC}(B)$, fixing membership), and the vote binding (keypers emit $\rho_{i,\cdot}$ under ctx_i with their consensus vote, Figure 1). The difference from TACET-P is only the mechanism: isolation reduces to pseudorandomness of the zero-mask, not signature unforgeability.

8.3 The combined noise budget

The one place the lattice port can silently break is the summed error: the core decryption noise, the flooding, and the masking error are all multiplied by recovery coefficients when t shares are combined, and must still land under the rounding threshold. We control this two ways. First, the flooding is weighted by the *low-norm* Pilvi recovery coefficients, so its summed width is at most $\kappa \sigma_{\text{fl}}$ with $\kappa^2 \geq \sum_{i \in S} \lambda_{i,S}^2$. Second, the context mask is the pseudorandom zero-sharing $z_i(\text{ctx}, j)$ of Section 8.1: it is built so that $\sum_{i \in S} \lambda_{i,S} z_i(\text{ctx}, j) = 0$ *exactly* (antisymmetric per-pair construction), so it contributes *no* residual error ($\beta_F = 0$) yet leaves each share pseudorandom and isolates contexts. (An almost-key-homomorphic-PRF realisation would contribute a bounded β_F instead; Remark 4.) The core decryption noise σ_{dec} is *not* amplified by κ , because the Pilvi coefficients reconstruct $\mathbf{s}$ exactly and that noise lives in the original ciphertext.

Lemma 1 (Combined opening error). *Let σ_{dec} bound the core decryption-noise width, σ_{fl} the flooding width, β_F the residual masking error ($\beta_F = 0$ for the exact zero-sharing), and let the Pilvi recovery coefficients satisfy $\sum_{i \in S} \lambda_{i,S}^2 \leq \kappa^2$. Then for any t -subset S each opening Δ_j equals $\mathbf{s}^\top \mathbf{u}_j$ plus an error of width at most*

$$\sigma_{\text{tot}} \leq \sqrt{\sigma_{\text{dec}}^2 + \kappa^2(\sigma_{\text{fl}}^2 + \beta_F^2)}.$$

Decapsulation is correct whenever $z \cdot \sigma_{\text{tot}} < q/4$ with z chosen so that $\text{erfc}(z/\sqrt{2}) \leq 2^{-\lambda}/N$.

The proof is the triangle inequality over independent (sub-)Gaussians plus the Pilvi coefficient bound; it is given in Section A. Section 8.4 shows the inequality holds with large margin at a concrete parameter set, and Section 10 confirms the predicted opening width $\kappa \sigma_{\text{fl}}$ empirically to three significant figures.

Remark 4 (On the masking error β_F). With the exact zero-sharing (Section 8.1) the masks cancel with no residual, so $\beta_F = 0$ and the budget is $\sigma_{\text{tot}} = \sqrt{\sigma_{\text{dec}}^2 + \kappa^2 \sigma_{\text{fl}}^2}$. This realisation uses per-pair keys and is exact for the participating

Quantity	Value	Source / constraint
Ring degree N	256	$X^N + 1$, NTT-friendly
Module rank k	6	LWE dimension $n = Nk = 1536$
Error width σ_e	3.2	short secret and error
Modulus q	$2^{31} - 511$	NTT prime, $q \equiv 1 \pmod{2N}$; budget closes with $z \approx 98$ margin
Committee (n_{kp}, t)	(13, 7)	honest-majority 7-of-13; replicated $\{0, 1\}$ -sharing
Recovery norm κ	≈ 2.65	replicated/CNF sharing, $\kappa = \sqrt{t}$ for every t -subset
Core noise σ_{dec}	$\approx 5.7 \cdot 10^2$	$\sqrt{2n} \sigma_e^2$ for $w - \mathbf{s}^\top \mathbf{u}$
Query bound Q	2^{12}	challenge-share observations (see Remark 5)
Flooding σ_{fl}	$\approx 2.1 \cdot 10^6$	Rényi: $\sigma_{\text{dec}} \sqrt{2\pi a Q N}$, $a = 2$ (the $\sqrt{N}$ is the ring correction)
Combined σ_{tot}	$\approx 5.5 \cdot 10^6$	Lemma 1, $\kappa = \sqrt{7}$
Decryption failure	$< 2^{-128}$	$z \cdot \sigma_{\text{tot}} < q/4$; realised $\leq 2^{-6998}$ (impl prime), $q/4 \approx 98 \sigma_{\text{tot}}$
Ciphertext ct	≈ 7 KB	$(k+1) \mathcal{R}_q$ elements $(\mathbf{u}, w)$, plus NIZK
Secret share $\mathbf{s}_i$	$\binom{n_{\text{kp}}-1}{t-1} = 924$ tuples	replicated $\{0, 1\}$ -sharing material per party at (13, 7)
Partial opening $\rho_{i,j}$	≈ 1 KB	one $\mathcal{R}_q$ element per ciphertext, plus π_i
Batch opening $\{\Delta_j\}$	$ B \cdot 1$ KB	$O(B)$, not constant (Remark 2; AUDIT Hole A)
PK MLWE hardness	136-bit (Q)	primal-uSVP at (n, q, σ_e) , confirmed by an estimator (Section 10); 142-bit at the impl prime

Table 4: A *sound, deployable* TACET-Q parameter set. The committee is 7-of-13 with the replicated $\{0, 1\}$ -sharing ($\kappa = \sqrt{t}$ for every subset), the one sharing we can *exhibit* that keeps the recovery norm low at scale; the noise budget (Lemma 1) and the simulation requirement (Remark 5) then close with vast margin ($q/4 \approx 98 \sigma_{\text{tot}}$) at the NTT prime $q = 2^{31} - 511$, and *every* random 7-of-13 committee decapsulates end to end (Section 10, `bench/deploy_validation.py`). A larger committee such as (100, 67) is *aspirational*: it would need a low-norm sharing with κ still small at $t = 67$, which no cited construction is known to deliver (the Pilvi sharing’s measured κ is exponential in t there, Section B and AUDIT §5), so its cost is the standing open problem, not a set we claim closes. Sizes exclude the NIZK proofs, which add ≈ 11 KB per ciphertext (measured, Section 10) and verify at mempool admission, off the decryption critical path. *Security note*: the 136-bit quantum figure is the primal-uSVP estimate; it is corroborated by a standalone estimator validated against Kyber/FrodoKEM (Section 10), with the dual attack not binding, but core-SVP is an optimistic model and a full lattice-estimator run with the gate-count model (which gives a higher figure) is still advisable before deployment.

committee. A realisation from an almost-key-homomorphic PRF with a shared zero key keeps the masks S -independent (any t -subset cancels) but contributes a bounded β_F ; achieving simultaneously a secure PRF, an $\mathcal{R}_q$ output, and $\beta_F = \Theta(\sigma_{\text{dec}})$ is delicate, since fine rounding (small β_F) weakens the PRF and coarse rounding (secure PRF) inflates β_F . The budget below leaves headroom for a nonzero β_F up to σ_{fl} scale without changing the modulus.

8.4 Concrete parameters

Table 4 fixes a parameter set and checks Lemma 1 and the simulation requirement together. The two constraints pull against each other: flooding forces q up, and a large q at small noise would weaken Module-LWE, so the dimension is raised to the FrodoKEM scale that Micciancio–Suhl identify. The set below closes the budget at $q \approx 2^{32}$ with a decryption-failure rate far below the 2^{-128} target. One implementation caveat: $q = 2^{32}$ is a power of two and is therefore *not* NTT-friendly (the negacyclic NTT needs $q \equiv 1 \pmod{2N}$). Our reference implementation uses the nearest prime $q \equiv 1 \pmod{512}$ below 2^{31} ($q = 2147483137$), at which the budget still closes with a realised failure rate $\approx 2^{-179}$ (a prime near 2^{32} gives $\approx 2^{-728}$); sizes change by under 4%.

8.5 Security

We state the results formally; proofs are in Section A. The first is the flooding lemma the composition rests on; it is the technical core, because the PILVI simulator must be redone for the masked, batched object rather than invoked as a black box.

Lemma 2 (Simulatable masked openings). *Let the ciphertext and flooding noises be discrete Gaussians with the widths of Table 4. There is a simulator that, given ek, the corrupt shares $\{\mathbf{s}_i\}_{i \in C}$ and per-pair masking keys, and the target keys, produces partial openings $\rho_{i,j}$ for honest i whose joint distribution, over up to Q queries under contexts containing the challenge with fewer than t honest openings each, has order- a Rényi divergence at most a constant $e^{1/2} \approx 1.65$ from the real openings (not $1 + o(1)$; Section A). By the multiplicative probability-preservation bound this preserves negligibility, so the masked openings leak no more than the keys they reveal.*

Remark 5 (The ring flooding correction: a $\sqrt{N}$ factor). The flooding width must be set with care in the *ring* setting. A partial opening $\rho_{i,j} \in \mathcal{R}_q$ is a full ring element, so the secret-dependent shift the flooding must hide is an N -dimensional vector of per-coordinate width σ_{dec} , i.e. of ℓ_2 norm $\sqrt{N} \sigma_{\text{dec}}$, and the Rényi divergence $\exp(a\pi \|\mu\|^2 / \sigma_{\text{fl}}^2)$ multiplies over all N coordinates and Q queries. The plain-LWE (scalar decryption) bound $\sigma_{\text{fl}} = \sigma_{\text{dec}} \sqrt{2\pi a Q}$ therefore *under-floods* by $\sqrt{N}$: at that width the total divergence is $\exp(a\pi N Q \sigma_{\text{dec}}^2 / \sigma_{\text{fl}}^2) = \exp(N/2) = \exp(128)$, so the masked openings would not hide the secret at all. The correct width is $\sigma_{\text{fl}} = \sigma_{\text{dec}} \sqrt{2\pi a Q N}$, restoring $D_a = e^{1/2}$. Keeping $Q = 2^{20}$ then forces $q \approx 2^{35}$ (and a larger dimension to hold 128-bit security); we instead read the query bound as $Q = 2^{12}$, at which $\sigma_{\text{fl}} = 2.1 \cdot 10^6$ exactly as in Table 4, the budget closes at $q \approx 2^{31}$, and the flooding is genuinely simulation-secure. This is confirmed empirically (closed form and Monte-Carlo) in Section 10; it was found by validating the simulator and is invisible to correctness tests, since under-flooding only *helps* decryption.

Theorem 4 (Post-quantum confidentiality). *In the QROM, for every quantum polynomial-time $\mathcal{A}$ making at most Q decryption queries,*

$$\text{Adv}_{\text{TACET-Q}, \mathcal{A}}^{\text{IND-CCA-CTX}}(\lambda) \leq \text{Adv}_{n, q, \sigma_e}^{\text{MLWE}}(\mathcal{B}_1) + \text{Adv}_{\text{zm}}^{\text{prf}}(\mathcal{B}_2) + \text{Adv}_{\text{NIZK}}^{\text{se}}(\mathcal{B}_3) + \varepsilon_{\text{fl}} + \text{negl}(\lambda),$$

where $\mathcal{B}_1, \mathcal{B}_2, \mathcal{B}_3$ are explicit reductions and the corruption bound is static $|C| \leq t - 1$. The flooding term ε_{fl} of Lemma 2 enters multiplicatively (via the order- a Rényi probability-preservation bound with constant divergence $e^{1/2}$), not as the clean additive summand the display suggests; the inequality is written additively for readability, with the precise coupling deferred to Section A. The $\text{Adv}_{\text{NIZK}}^{\text{se}}$ term carries the QROM loss of the straight-line extractor, and Adv^{prf} is the pseudorandomness of the context zero-mask.

Proof sketch. A sequence of games: (1) simulate every NIZK in Encap answers and straight-line extract witnesses from Decap queries, so the decryption oracle is answered without key shares, at cost $\text{Adv}_{\text{NIZK}}^{\text{se}}$ in the QROM; (2) replace honest partial openings by Lemma 2’s simulator, at cost ε_{fl} ; (3) replace the zero-masks of non-challenge contexts by random, at cost Adv^{prf} ; (4) the residual distinguisher on K^* becomes a Module-LWE distinguisher on $(\mathbf{A}, \mathbf{b}, \mathbf{u}^*)$, at cost Adv^{MLWE} . Full hybrids in Section A. $\square$

Theorem 5 (Context isolation). *TACET-Q satisfies Definition 3 in the QROM: openings under any $\text{ctx}' \neq \text{ctx}$ are simulatable from public data, reducing to pseudorandomness of the context zero-mask at the inputs $(\text{ctx}, \cdot)$ that no honest party evaluated. Hence the per-context counting in Definition 2 is the only restriction.*

Theorem 6 (Rogue ciphertexts and robustness). *TACET-Q satisfies Definition 4: Verify rejects malformed ciphertexts before batching; membership gating means a well-formed adversarial $\text{ct}^\dagger$ yields no cross-leakage; and a share ρ_i with an invalid π_i is rejected with its index identified, while any t valid shares reconstruct the correct Δ . Soundness of the well-formedness and share proofs reduces to Module-SIS.*

8.6 What remains open

Resolved in the literature (silent post-quantum setup). A *silent* t -of- n post-quantum threshold KEM — joint key a deterministic public function of locally generated keys, no DKG — is now achievable *from a falsifiable lattice assumption*. Distributed monotone-policy encryption (Champion–Wu, EUROCRYPT 2026) directly generalises threshold encryption with silent setup and is built from *decomposed* LWE; its foundation, *distributed broadcast encryption* (Champion–Wu, TCC 2024), is the $t=1$ case from ℓ -succinct LWE (Wee, CRYPTO 2024). Both assumptions are falsifiable (unlike the evasive-LWE/witness-encryption route that gave the earlier, selectively secure $t=1$ object). We implement and validate the silent mechanism — silent per-user key generation, set/threshold encryption, t -of- n decryption, sub-threshold hiding — in `silentpq/` (see `docs/silent-pq.md`). Notably, the threshold layer requires the secret-sharing of the target to have *small (binary) reconstruction coefficients*: Shamir’s large coefficients break the modulus, the *same* tension this paper meets at κ (Section B, `AUDIT.md` §5). We further build the *constant-size* version: the DMPE three-element ciphertext, independent of the committee, on top of the genuine WW25 sparse *matrix commitment* `Com/Ver/Open` (`silentpq/wee_commitment.py`, where the defining relation $\mathbf{Cv}_{L,i} = \mathbf{m}_i - \mathbf{Bz}_i$ and the gadget identity are reconstructed and checked exactly); the full threshold runs on this commitment, and a sweep confirms the ciphertext stays flat as the committee grows ($864\times$ smaller than per-set broadcast at $n_{\text{kp}}=13, t=7$; `bench/succinct_validation.py`). The remaining cost over TACET-Q’s MLWE is the stronger (still falsifiable) ℓ -succinct/decomposed-LWE assumption, selective security, and the asymptotic (non-concretely-small) commitment parameters. A threshold IND-CCA statement and structured proof for this silent scheme (in the QROM, reducing to the cited CPA / FO-QROM / smudging inputs, with the hop composition and noise budget machine-checked) is Theorem 7. TACET-Q can adopt this in place of its one-time DKG to drop the D2 debit.

The other open items are narrower: fully *adaptive* corruptions for the batched lattice scheme (the static bound is inherited from the core, Remark 3); removing the a priori bound Q while keeping a polynomial modulus; and compressing the lattice ciphertext toward the pairing sizes. None blocks deployment.

Migration. Because TACET-P and TACET-Q share the syntax of Definition 1 and the same anchor semantics, a deployment can run them side by side, encapsulating each transaction under both and AEAD-binding the two keys, so confidentiality holds as long as *either* the pairing or the lattice assumption stands. Unlike a pure harvest-now-decrypt-later hedge, TACET-Q keeps the full threshold property under a quantum adversary, not only secrecy of contents.

9 Distributed key generation

TACET-Q is transparent but not silent: it needs a one-time setup that hands each keyper a low-norm share of $\mathbf{s}$ and establishes the per-pair masking keys $\{\kappa_{i,i'}\}$ of the context zero-sharing. The subtlety is in the share of $\mathbf{s}$: these cannot be arbitrary shares. The core’s polynomial modulus depends on the recovery coefficients being low-norm (Lemma 1), so the DKG must output shares in the *Pilvi* sharing, not the additive or Shamir-over-the-integers shares a generic lattice DKG produces. Clearing Shamir denominators reintroduces the $(n!)^2$ blow-up the polynomial modulus was chosen to avoid — a hazard we confirm empirically in Section 10, where generic Shamir recovery at $(100, 67)$ has norm $\approx 5.4 \cdot 10^9$ and breaks decryption.

Protocol. Each keyper samples a short additive contribution to $\mathbf{s}$, publishes a Module-SIS commitment to it with a lattice NIZK of shortness, and verifiably reshapes it into the *Pilvi* structure; the joint secret is the sum, so no party learns $\mathbf{s}$ (dealerless). The per-pair masking keys are established by a one-round pairwise key agreement. This is one round of broadcast plus pairwise shares, $O(n_{\text{kp}}^2)$ communication, run once at committee formation and amortised over every future block. It instantiates from the Ring-LWE dealerless DKG line; adapting that output to the low-norm *Pilvi* sharing, and proving the resulting shares meet Lemma 1’s norm bound with the claimed κ , is the one piece of the setup that needs its own short treatment, and we flag it as such.

Rotation. Committee changes use proactive resharing of $\mathbf{s}$ and re-keying of the masking pairs. Under static corruptions this is a re-run of the DKG; an adaptive-corruption refresh is tied to the open adaptivity question (Remark 3).

10 Reference implementation and benchmarks

We implemented TACET-Q as a modular reference library (Python/NumPy): negacyclic ring arithmetic with a number-theoretic transform, the Module-LWE Lindner–Peikert core, low-norm threshold sharing, the context-keyed pseudorandom zero-mask of Section 8.1, a Merkle vector commitment, discrete-Gaussian flooding, the dealerless DKG, and the full Setup/Encap/Verify/PartDec/Open/Decap pipeline, plus a *real* zero-knowledge NIZK of well-formedness (below). One thing is abstracted at *committee scale*: *PILVI*’s low-norm t -of- n sharing. The load-bearing property is that the recovery norm κ stays small for the opening committee; we verify this property is exactly realisable for *every* t -subset (not just a fixed one) by a replicated/CNF sharing (`tacet.sharing.ReplicatedLowNorm`), whose reconstruction coefficients are all in $\{0, 1\}$, giving $\kappa = \sqrt{t}$ for any committee — checked exhaustively over all t -subsets and shown to close the budget for any committee (`bench/sharing_lownorm.py`). Its cost is $\binom{n-1}{t-1}$ ring elements per party, so at $(100, 67)$ it is replaced by the polynomial-size constructions that preserve the same $\{0, 1\}$ -recovery property (Boudgoust–Scholl; Micciancio–Suhl; *PILVI*); reimplementing those at scale is the one piece we cite rather than build. An exact pairwise zero-sharing is used for the context mask. The code, tests and benchmark scripts accompany this paper.

The well-formedness NIZK is real, not abstracted. Encap attaches and Verify checks a Fiat–Shamir-with-aborts proof (TACET.NIZK, in the Lyubashevsky/LNP style with uniform masking) of knowledge of short $(\mathbf{r}, \mathbf{e}_1, \mathbf{e}_2)$ with $\mathbf{A}^\top \mathbf{r} + \mathbf{e}_1 = \mathbf{u}$ and w consistent, binding $\mathbf{ad}$ into the challenge. It is sound (a rogue $\mathbf{u}$ with no short preimage cannot be proven; reduces to MSIS/ISIS), zero-knowledge (the response is uniform in the masking box, independent of the witness — verified to match the simulator), and non-interactive. Measured: ≈ 11 KB, prove 0.07 s, verify 0.04 s (Table 5). What it does not add is LaBRADOR-style succinctness or a binary- μ proof (the latter affects only a malicious encryptor’s own key, not others’ security). We probed the succinct route directly: the LNP authors’ LAZER library (LaBRADOR) builds and runs in our environment and its linear-relation prover maps exactly onto this statement, but a LaBRADOR proof at our degree/modulus is ≈ 22 KB — *larger* than the 11 KB hand-rolled

proof, because its ~ 20 KB constant overhead dominates at single-ciphertext scale. Succinctness therefore pays only when *aggregating* a whole block’s $|B|$ ciphertexts into one proof, not per ciphertext: amortising that fixed overhead, a single batch proof undercuts the $|B| \times 11$ KB hand-rolled cost from $|B| \geq 3$ on (a conservative linear-content estimate; $\geq 3 \times$ smaller at $|B|=16$, `lazer/batch_projection.py`). Producing the actual TACET batch proof is gated only on a Sage-version fix to LAZER’s parameter codegen, not on the construction.

Validated properties. A test suite (24 tests, all passing) exercises: end-to-end correctness; threshold reconstruction by any t -subset (structurally) and end-to-end opening on a low-norm committee; that $t - 1$ openings cannot recover a key even when force-combined; *context isolation* — a t -set mixing two forks does not open the block, and the zero-mask cancels under one context but not when mixed; only-included opening; NIZK completeness, soundness (ad/ciphertext tamper) and zero-knowledge; identifiable abort on a bad share; that a well-formed adversarial ciphertext does not corrupt honest openings; dealerless DKG correctness; and the flooding-divergence relations of Remark 5.

Noise budget, empirically. At the parameters of Table 4 (NTT prime $q = 2^{31} - 511$) the measured opening-noise width is $3.38 \cdot 10^7$, matching the analytic $\kappa \sigma_{\text{fl}} = 3.37 \cdot 10^7$ (Lemma 1) to three significant figures; across trials the noise never exceeds 27% of $q/4$, implying a decryption-failure rate $\leq 2^{-179}$, comfortably under the 2^{-128} target. As a control, a *generic* Shamir sharing at $(100, 67)$ has recovery norm $\approx 5.4 \cdot 10^9$ (blow-up $\approx 3 \cdot 10^8$), its flooded noise exceeds $q/4$, and decryption fails — a direct, empirical confirmation that the low-norm sharing is load-bearing, not cosmetic.

GPU-scale validation of the noise budget. The trial-based check above sees only a few thousand noise coordinates. To stress the *tail* — which is what the erfc failure bound rests on — we resample the combined opening noise on a GPU (`bench/gpu_validation.py`, an RTX 4090) at $2 \cdot 10^9$ samples per sampler, about $2.6 \cdot 10^5 \times$ more coordinates, faithfully reproducing the model noise $= \text{core}_{\text{dec}} - \sum_{i \in S} \lambda_i \text{Flood}_i$ with the realised low-norm recovery coefficients. For each of the three realised flooding samplers (rounded Gaussian, Bates, and the constant-time discrete Gaussian of the native build), the empirical width matches σ_{tot} (evaluated at the realised $\kappa = 16.28$) to five significant figures; the combined noise is Gaussian, its two-sided tail tracking $\text{erfc}(c/\sqrt{2})$ to $\sim 0.1\%$ out to 5σ and staying within sampling noise to $\sim 6\sigma$; and the largest deviation seen across $2 \cdot 10^9$ draws is only $\approx 40\%$ of $q/4$ (about 6.3σ). The implied decryption-failure rate is $\leq 2^{-182}$, confirming the $\geq 2^{-128}$ margin holds far into the tail, not just on average.

Concrete-security estimate (real estimator). We ran the community-standard lattice estimator (Albrecht et al., under SageMath) on the TACET-Q instance ($n = 1536, \sigma_e = 3.2$). At $q = 2^{32}$ the binding attack is primal-uSVP at **150-bit** classical core-SVP (ADPS16), with the dual and dual-hybrid attacks at 152.7 (non-binding); under the modern MATZOV gate-count model the figure rises to **176-bit**. At the implementation prime $2^{31} - 511$ these become 157-bit (core-SVP) and 183-bit (MATZOV), the smaller modulus being slightly harder. The quantum core-SVP figure is $0.265\beta \approx 136$ -bit at $\beta = 514$. A self-contained estimator we also provide (validated to reproduce Kyber-512/768 exactly) agrees. Thus 128-bit (quantum, core-SVP) holds with margin, and the conservative gate model is far above it.

Machine-checked core (Coq). Three machine-checked developments (Coq, proofs/, standard library only — they reduce to the usual classical-reals and functional-extensionality axioms, with no `Admitted` and no added axioms) cover the algebraic hearts of the steps Section A flags as subtle. **(i) Noise composition** (`tacet_noise.v`): the corrected flooding variance is exactly $N \times$ the draft’s; the draft width gives total Rényi exponent $N/2$ ($= 128$ at $N=256$) while the corrected width gives $1/2$; the combined opening variance is monotone (the worst-case set is safe); and the exact zero-sharing ($\beta_F=0$) is noise-optimal. **(ii) Context isolation** (`tacet_isolation.v`): the recovery-weighted context mask is the sum, over the ordered pairs of the participating set, of an *antisymmetric* matrix, which is identically 0 — so the mask cancels exactly under one context with no residual noise (the $\beta_F=0$ fact Lemma 1 relies on, and the cancellation underlying Theorem 5); cancellation is shown equivalent to pairwise context agreement, the formal sense in which mixing forks fails to cancel. **(iii) The flooding hop** (`tacet_renyi.v`): finite Cauchy–Schwarz is checked from a sum-of-squares, and from it the order-2 probability-preservation bound $\Pr_{\text{real}}[E] \leq \sqrt{D_2 \cdot \Pr_{\text{sim}}[E]}$ — exactly the multiplicative $a=2$ step Theorem 4 uses in place of a naive additive ε_{fl} . **(iv) Advantage assembly and q_{min}** (`tacet_qrom.v`): the hybrid telescopes — $|\Pr[G_0] - \frac{1}{2}|$ is at most the sum of the four hop bounds (triangle inequality along the chain) — and the budget “closes” iff $q > 4\kappa \sigma_{\text{tot}}$, the q_{min} of the parameter table. We take the single-coordinate Gaussian/Rényi facts and the per-hop reductions as inputs (the former standard, the latter on paper); a full machine-checked QROM IND-CCA proof — the measure-and-reprogram extractor in particular — remains external work.

Object / operation	Measured (PAPER set, $ B = 16$)	
ring element $\mathcal{R}_q$	packed	0.97 KB
public key $\mathbf{b}$ (k elts)	+32 B seed for $\mathbf{A}$	5.81 KB
ciphertext $\text{ct} = (\mathbf{u}, w)$	$(k+1)$ elts, +NIZK off-path	6.78 KB
secret share s_i	$\mathcal{R}_q^k$	5.81 KB
opening $\rho_{i,j}$ (per ct)	one $\mathcal{R}_q$ element	0.97 KB
batch opening $\{\Delta_j\}$	$O(B)$, $ B = 16$	15.5 KB
KGen (low-norm)	one-time	151 ms
Encap	per ciphertext	38.6 ms
Verify	at mempool admission, off critical path	39.6 ms
PartDec (per keyper, $ B = 16$)	18.8 ms/ct	301 ms
Open ($t = 67$, $ B = 16$)	2.0 ms/ct	31.6 ms
Decap (per ciphertext)	on the critical path	0.14 ms

Table 5: Reference-implementation sizes and single-thread wall-clock timings at the Table 4 parameters (Python/NumPy; a native build would be far faster). Sizes track the analytic table; the batch opening is $O(|B|)$ per the correction of Remark 2. Decap, the only step on the block’s critical path, is sub-millisecond per ciphertext.

Flooding simulator (validating lem:flood). We verify Remark 5 both in closed form and by Monte-Carlo: the per-coordinate order-2 Rényi divergence matches $\exp(\delta^2/\sigma_{\text{fl}}^2)$; multiplied over the N ring coordinates and Q queries it is $\exp(N/2) = \exp(128)$ at the plain-LWE width (the masked openings would leak), and $e^{1/2} \approx 1.65$ at the corrected width $\sigma_{\text{dec}}\sqrt{2\pi aQN}$ with $Q = 2^{12}$. We also confirm the simulator is faithful: openings centred at the public target plus flooding, produced *without* the secret shares, decode to the correct keys.

Which flooding sampler the Rényi bound actually permits. Lemma 2 needs the per-coordinate divergence to be *finite*, and this is sharper than matching the width. Computing the order-2 divergence of each realised sampler exactly (bench/gpu_validation.py) shows that a full-support discrete (or rounded) Gaussian gives $R_2 = \exp(\delta^2/\sigma^2)$ to machine precision, but the constant-time *Bates* sampler (a sum of 12 uniforms), being supported on the compact interval $[-6\sigma, 6\sigma]$, places mass under a shift where the unshifted law has none — so its Rényi divergence is *infinite* for every $\delta > 0$, and the clean Rényi flooding bound does not apply to it. It is nonetheless safe at TACET-Q’s parameters by a *statistical-distance* argument: the offending edge mass is $(\delta/\sigma)^{12}/12!$, which at the scheme’s per-coordinate shift $\delta/\sigma_{\text{fl}} = \sigma_{\text{dec}}/\sigma_{\text{fl}} \approx 2.8 \cdot 10^{-4}$ is $\approx 2^{-171}$ per coordinate and $\leq 2^{-151}$ over all NQ coordinate-queries — below 2^{-128} , but secured by the wrong tool. The native build therefore uses a *full-support* constant-time discrete Gaussian (a Micciancio–Walter convolution of a branch-free base sampler, rust/src/sampling.rs), whose measured R_2 reproduces the Gaussian closed form (1.287 vs. 1.284 at the $\delta = \sigma/2$ stress shift) and whose tail is genuinely Gaussian, so Lemma 2 applies directly. This is a subtlety only the exact (not Monte-Carlo) divergence exposes: a fixed-sample Rényi estimate of the Bates sampler returns a finite, innocuous-looking ≈ 1.3 .

Constant-time native build (Rust). We also provide a side-channel-conscious native port (rust/) that mirrors the reference algorithm-for-algorithm but writes every secret-dependent operation — Barrett modular arithmetic, the negacyclic NTT, the message decode, share combination, and the share-validity tag check — branch-free, with no value-dependent memory addressing. Two complementary checks back the claim. *Control flow*: under ctgrind (secret buffers poisoned and the secret-dependent code run under valgrind) the build reports ERROR SUMMARY: 0 errors — no secret-dependent branch or memory access, the strong environment-independent guarantee. *Timing*: a dudect fixed-vs-random Welch t -test over $\sim 10^7$ measurements (one pinned core) leaves the branch-free decode, modular multiply and NTT multiply all below the $|t| < 4.5$ no-leakage threshold (decode arithmetic stable at $|t| \approx 2.1$ – 2.5); isolating the message-decode *arithmetic* from its output-buffer allocation shows the small residual is allocator/cache state, not a secret-dependent path. Every deterministic primitive (SHAKE domain separation, the KDF, the random-oracle ring expansion, the NTT, the Merkle commitment, the NIZK challenge) is pinned byte-for-byte to the Python reference by known-answer vectors, and a canonical bit-packed wire format with a panic-free, range-checking parser is fuzzed (in-process and via cargo-fuzz). At the Table 4 parameters the native Encap/Verify/Decap run in 2.5/0.73/0.08 ms — roughly $15\times$ the NumPy reference — and reproduce the analytic noise budget of the Python reference (the $\sqrt{N}$ -corrected flooding $\sigma_{\text{fl}} = 2.06 \cdot 10^6$; the combined σ_{tot} and decryption-failure rate track Table 4). This closes the “constant-time native build” item below; what a deployment still wants is a vetted production discrete-Gaussian sampler (the build ships one; see above) and a like-for-like head-to-head against BEAST-mev’s optimised pairing code.

11 Integration with Ethereum

The encrypted-mempool EIP under discussion is deliberately encryption-scheme agnostic, so the contract is precise: produce a per-block opening from keyper material with no extra latency round. The points that decide whether TACET-Q actually drops in:

- **Anchor versus block identity.** ctx must be fixed when keypers act, but the final block may not be. We set $\text{ctx} = (\text{slot}, \text{proposer commitment}, \text{VC}(B))$ where B is the ordered list of encrypted transactions the proposer commits to. All three are available at proposal time, which is the proposer-commitment hook the EIP already exposes.
- **Reorgs and equivocation.** A keyper emits at most one ρ_i per slot, bound to one ctx . Shares for a reorged-out proposal bind a different ctx and are inert against the canonical block by Theorem 5, so a reorg cannot retroactively open the surviving block from losing-fork shares, and sub-threshold openings of an abandoned block leak nothing. An equivocating proposer who publishes two ctx for one slot is caught by the one-share-per-slot rule and slashed.
- **Liveness fallback.** If fewer than t keypers are live, the batch does not open. The safe behaviour is to leave those transactions encrypted and re-propose them, trading liveness of *reveal* for confidentiality; the alternative, a timeout fallback that reveals without the committee, reintroduces the front-running window. We take the former and note the censorship and free-option tension this leaves, which is a property of encrypted mempools generally, not of TACET-Q.
- **Accountability that is actionable.** The share-verifiability proof π_i gives identifiable abort (Theorem 6): a keyper that publishes a malformed share is identified and slashable on chain. This is distinct from tracing *within-protocol* collusion, which TACET-Q does not provide (Section 12).
- **Committee model.** Shutter keypers are a committee separate from the validators, so the vote-piggyback binds keyper consensus, not validator consensus; the proposer commitment is the bridge between the two. The free-option problem a proposer enjoys by choosing whether to finalise is orthogonal and is the target of reorder-constraining proposals such as EIP-7793.

12 Limitations and status

We separate what this paper establishes from what a deployment still requires, because the gap is the honest content of a preprint at this stage.

What still needs external work.

1. **Production implementation and a Rust comparison (native build done).** Section 10 gives a reference implementation (Python/NumPy) and a constant-time native port (rust/) that is ctgrind-clean, dduct-checked, KAT-locked to the reference, fuzzed, and ships a full-support constant-time discrete-Gaussian flooding sampler. What remains is not the build itself but a like-for-like head-to-head against BEAST-mev’s optimised pairing code under a 12-second slot, and an independent audit. A *succinct* NIZK turns out not to help per ciphertext (LaBRADOR is ≈ 22 KB vs our 11 KB at this scale, Section 10); the worthwhile follow-up is instead a single *batch-aggregated* well-formedness proof via LaBRADOR.
2. **Estimator confirmation (done).** The community-standard lattice estimator confirms 150-bit classical core-SVP / 176-bit MATZOV-gate at $q = 2^{32}$ (136-bit quantum core-SVP), with the dual attack non-binding (Section 10). What remains is a deployment-grade re-tune (e.g. shaving margin) and tracking estimator updates, not a yes/no security question.
3. **Proof verification (partial).** We give the reductions (Theorems 4 to 6) and the flooding lemma (Lemma 2), with hybrids in Section A; the noise-budget and $\sqrt{N}$ flooding composition, the exact zero-mask cancellation behind context isolation, and the order-2 probability-preservation step of the flooding hop are now machine-checked in Coq (Section 10). What remains is the harder part: a machine-checked *QROM IND-CCA* proof (the measure-and-reprogram extractor and the multiplicative Rényi coupling), which is at the edge of current proof-assistant tooling and is genuine external work.

Known weaknesses, by design.

1. **Static corruptions** (Remark 3); fully adaptive batched security is **Open** for both instantiations.
2. **Not silent post-quantum** (D2): TACET-Q needs the one-time DKG of Section 9, because silent post-quantum threshold decryption is **Open** (Section 8.6).

3. **Larger ciphertexts** (D11): FrodoKEM-scale, the price of lattices.
4. **Linear batch opening** (D13): the per-block opening is $O(|B|)$ ring elements, not constant. Unlike the pairing line, the linear lattice route has no succinct-opening structure (Remark 2); a SNARK over the openings would have to be re-checked against the BLT barrier.
5. **A priori query bound** Q from the polynomial-modulus analysis; mild for finite per-block batches, but removing it at a small modulus is open.
6. **Large per-party material** (D10); $O(n_{kp})$ DKG and verification data.
7. **No collusion tracing**. TACET identifies malformed shares but does not trace keypers who collude within the protocol; the traceable-threshold and snitching lines (Bormet et al.; Dziembowski et al.) are complementary and unanalysed here.

Acknowledgements

This note responds to Wagner’s encrypted-mempool literature note and builds entirely on the cited work of the silent-setup, batched, context-dependent and lattice threshold-encryption lines. Errors are the author’s.

References

- [1] B. Wagner. *Encrypted Mempool / Threshold Encryption Literature*. Ethereum HackMD note, 2026.
- [2] S. Garg, D. Kolonelos, G.-V. Policharla, M. Wang. *Threshold Encryption with Silent Setup*. CRYPTO 2024. ePrint 2024/263.
- [3] J. Bormet, S. Faust, H. Othman, Z. Qu. *BEAT-MEV: Epochless Approach to Batched Threshold Encryption for MEV Prevention*. USENIX Security 2025. ePrint 2024/1533.
- [4] J. Bormet, A. R. Choudhuri, S. Faust, S. Garg, H. Othman, G.-V. Policharla, Z. Qu, M. Wang. *BEAST-MEV: Batched Threshold Encryption with Silent Setup for MEV Prevention*. ePrint 2025/1419.
- [5] D. Boneh, B. Bünz, K. Nayak, L. Rotem, V. Shoup. *Context-Dependent Threshold Decryption and its Applications*. ASIACRYPT 2025. ePrint 2025/279.
- [6] C. Brzuska, M. Klooß, I. K. Y. Woo. *Threshold Public-Key Encryption: Definitions, Relations, and CPA-to-CCA Transforms*. ePrint 2025/1665.
- [7] D. Boneh, E. Laufer, E. N. Tas. *Batch Decryption without Epochs and its Application to Encrypted Mempools*. ePrint 2025/1254.
- [8] A. R. Choudhuri, S. Garg, J. Piet, G.-V. Policharla. *Mempool Privacy via Batched Threshold Encryption: Attacks and Defenses*. ePrint 2024/669.
- [9] A. R. Choudhuri, S. Garg, G.-V. Policharla, M. Wang. *Practical Mempool Privacy via One-Time Setup Batched Threshold Encryption*. USENIX Security 2025.
- [10] Agarwal et al. *Weighted Batched Threshold Encryption with Applications to Mempool Privacy*. ePrint 2025/2115.
- [11] S. Garg, A. Jain, P. Mukherjee, R. Sinha, M. Wang, Y. Zhang. *hinTS: Threshold Signatures with Silent Setup*. IEEE S&P 2024. ePrint 2023/567.
- [12] J. Champion, D. J. Wu. *Distributed Monotone-Policy Encryption for DNFs from Lattices*. EUROCRYPT 2026. ePrint 2026/318.
- [13] *Distributed-policy threshold encryption from ℓ -decomposed LWE* (selective, static, transparent setup, per Wagner’s note). ePrint 2026/372.
- [14] B. Waters, D. J. Wu. *Silent Threshold Cryptography from Pairings: Expressive Policies in the Plain Model*. EUROCRYPT 2026. ePrint 2025/1547.
- [15] J. Gong, B. Waters, H. Wee, D. J. Wu. *Threshold Batched Identity-Based Encryption from Pairings in the Plain Model*. ePrint 2025/2103.
- [16] R. Bhattacharyya, J. Bormet, S. Faust, P. Mukherjee, H. Othman. *CCA-secure Traceable Threshold (ID-based) Encryption and Application*. ePrint 2025/341.
- [17] Z. Brakerski, V. Vaikuntanathan. *Constrained Key-Homomorphic PRFs from Standard Lattice Assumptions*. TCC 2015.
- [18] V. Cini, R. W. F. Lai, I. K. Y. Woo. *Pilvi: Lattice Threshold PKE with Small Decryption Shares and Improved Security*. ASIACRYPT 2025. ePrint 2025/1691.
- [19] K. Boudgoust, P. Scholl. *Simple Threshold (Fully Homomorphic) Encryption from LWE with Polynomial Modulus*. ASIACRYPT 2023. ePrint 2023/016.
- [20] D. Micciancio, A. Suhl. *Simulation-Secure Threshold PKE from LWE with Polynomial Modulus*. IACR Communications in Cryptology, 2025. ePrint 2023/1728.

- [21] R. Bendlin, I. Damgård. *Threshold Decryption and Zero-Knowledge Proofs for Lattice-Based Cryptosystems*. TCC 2010.
- [22] D. Boneh, R. Gennaro, S. Goldfeder, A. Jain, S. Kim, P. M. R. Rasmussen, A. Sahai. *Threshold Cryptosystems from Threshold Fully Homomorphic Encryption*. CRYPTO 2018.
- [23] A. Passelègue, D. Stehlé. *Low Communication Threshold Fully Homomorphic Encryption*. ASIACRYPT 2024.
- [24] V. Lyubashevsky, N. K. Nguyen, M. Plançon. *Lattice-Based Zero-Knowledge Proofs and Applications: Shorter, Simpler, and More General*. CRYPTO 2022. ePrint 2022/284.
- [25] W. Beullens, G. Seiler. *LaBRADOR: Compact Proofs for R1CS from Module-SIS*. CRYPTO 2023.
- [26] S. Katsumata. *A New Simple Technique to Bootstrap Various Lattice Zero-Knowledge Proofs to QROM Secure NIZKs*. CRYPTO 2021. ePrint 2021/927.
- [27] J. Bos, C. Costello, L. Ducas, I. Mironov, M. Naehrig, V. Nikolaenko, A. Raghunathan, D. Stebila. *Frodo: Take off the Ring! Practical, Quantum-Secure Key Exchange from LWE*. ACM CCS 2016.
- [28] D. Catalano, D. Fiore. *Vector Commitments and Their Applications*. PKC 2013.
- [29] M. R. Albrecht, R. Player, S. Scott. *On the Concrete Hardness of Learning with Errors*. Journal of Mathematical Cryptology, 2015. ePrint 2015/046.

A Proof of the main theorems

These are structured reductions and proof sketches. The hop costs are explicit, but the concrete QROM constants and the masked-opening simulation are at the level a referee or a proof assistant should still check; Section 12 flags this.

A.1 Combined opening error (Lemma 1)

Fix a position j and write $\Delta_j = \sum_{i \in S} \lambda_{i,S} \rho_{i,j}$, substituting $\rho_{i,j} = \mathbf{s}_i^\top \mathbf{u}_j + z_i(\text{ctx}, j) + \text{Flood}_{i,j}$ from Construction 2. Linearity of the Pilvi sharing gives $\sum_i \lambda_{i,S} \mathbf{s}_i = \mathbf{s}$, so the core part collapses to the intended value $\mathbf{s}^\top \mathbf{u}_j$. The zero-mask cancels by construction: $\sum_i \lambda_{i,S} z_i(\text{ctx}, j) = 0$ exactly for the exact zero-sharing (residual η with $\|\eta\| \leq \kappa \beta_F$ for an almost-KH-PRF realisation; $\beta_F = 0$ here). The flooding terms sum to a discrete Gaussian of width at most $\kappa \sigma_{\text{fl}}$ with $\kappa^2 \geq \sum_i \lambda_{i,S}^2$, and the core decryption noise (which is *not* amplified by κ , living in the original ciphertext) has width σ_{dec} . These contributions are independent, so by sub-Gaussian addition the total error has width $\sigma_{\text{tot}} \leq \sqrt{\sigma_{\text{dec}}^2 + \kappa^2(\sigma_{\text{fl}}^2 + \beta_F^2)}$. Each of the N coefficients of Δ rounds correctly when its error is below $q/4$; a Gaussian tail bound gives per-coefficient failure $\text{erfc}(z/\sqrt{2})$ at threshold $z\sigma_{\text{tot}}$, and a union bound over N coefficients gives overall failure $\leq N \cdot \text{erfc}(z/\sqrt{2})$, which is $\leq 2^{-\lambda}$ for the stated z . $\square$

A.2 Simulatable masked openings (Lemma 2)

The simulator $\mathcal{S}$, on an honest query (i, ctx, B) , outputs $\tilde{\rho}_{i,j} = v_{i,j} + \text{Flood}_{i,j}$ where $v_{i,j}$ is the public-combiner value it can compute from $\mathbf{u}_j$ and the simulated target and $\text{Flood}_{i,j}$ is a fresh discrete Gaussian of per-coordinate width σ_{fl} . The real opening is centred at $v_{i,j} + \mu_{i,j}$ where $\mu_{i,j} \in \mathcal{R}_q$ is the secret-dependent shift: a *full ring element* whose N coordinates each have width $\Theta(\sigma_{\text{dec}})$, so $\|\mu_{i,j}\|_2 \leq B_{\text{shift}}$ with $B_{\text{shift}} = \Theta(\sqrt{N} \sigma_{\text{dec}})$ (this $\sqrt{N}$ is the ring correction of Remark 5; the per-coordinate framing that drops it is the error a naive port makes). For discrete Gaussians of equal width the Rényi divergence of order a is $D_a \leq \exp(a\pi \|\mu_{i,j}\|^2 / \sigma_{\text{fl}}^2)$ (up to a smoothing term), and Rényi divergence is multiplicative over the Q independent queries, so $D_a(\text{real}^Q \parallel \text{sim}^Q) \leq \exp(a\pi Q N \sigma_{\text{dec}}^2 / \sigma_{\text{fl}}^2)$. With $\sigma_{\text{fl}} = \sigma_{\text{dec}} \sqrt{2\pi a Q N}$ and $Q = 2^{12}$ as in Table 4, the total exponent is the *constant* $\frac{1}{2}$ and $D_a \leq e^{1/2} \approx 1.65$ — a constant bounded away from 1, *not* $1 + o(1)$. (Forcing $1 + o(1)$ would need an extra $\omega(1)$ factor in σ_{fl} , raising q ; we keep the smaller modulus and the constant divergence.) The correct probability-preservation step is then *multiplicative*, not a clean additive hop: for any event E , $\Pr_{\text{real}}[E] \leq (D_a \cdot \Pr_{\text{sim}}[E])^{(a-1)/a}$, which preserves negligibility but couples to the rest of the hybrid. For the IND *decision* game this is the Bai–Langlois–Lepoint–Stehlé–Steinfeld / Micciancio–Suhl treatment rather than a black-box additive ε_{fl} ; Theorem 4 reads its flooding term in that sense. This is the step where the PILVI analysis is redone for the masked, batched object, and is exactly what we flag for independent verification. $\square$

A.3 Post-quantum confidentiality (Theorem 4)

A hybrid over four games; $\Pr[G_i]$ is the probability $\mathcal{A}$ outputs $b' = b$ in G_i .

- G_0 : the real $\text{Exp}^{\text{IND-CCA-CTX}}$.
- $G_0 \rightarrow G_1$: simulate the NIZK proofs attached by Encap and straight-line extract the witness $(\mathbf{r}, \mathbf{e}_1, \mathbf{e}_2)$ from each Decap query, using the QROM simulation-extractable proof system (Katsumata transform). Decap is then answered by recomputing w from the extracted witness, with no key shares. $|\Pr[G_1] - \Pr[G_0]| \leq \text{Adv}_{\text{NIZK}}^{\text{se}}(\mathcal{B}_3)$, which carries the QROM extractor loss.
- $G_1 \rightarrow G_2$: answer every honest $\mathcal{O}_{\text{PartDec}}$ with the simulator of Lemma 2. By admissibility, every context containing ct^* has fewer than t honest openings, so the simulator is never asked to exceed the threshold. $|\Pr[G_2] - \Pr[G_1]| \leq \epsilon_{\text{fl}}$.
- $G_2 \rightarrow G_3$: replace the honest zero-mask contributions $z_i(\text{ctx}', \cdot)$ by uniform for every context ctx' other than the challenge context. Under $< t$ corrupt parties the per-pair keys touching an honest party are hidden, so this is sound by pseudorandomness of the zero-mask: $|\Pr[G_3] - \Pr[G_2]| \leq \text{Adv}^{\text{prf}}(\mathcal{B}_2)$.
- $G_3 \rightarrow G_4$: the challenge key is $K^* = G(\lfloor w^* \rfloor, \text{ad}^*)$ with $w^* = \mathbf{b}^\top \mathbf{r}^* + e_2^*$. Embed a Module-LWE instance into $(\mathbf{A}, \mathbf{b})$ and the challenge $\mathbf{u}^*$; a distinguisher between the real K^* and random becomes an MLWE distinguisher. $|\Pr[G_4] - \frac{1}{2}| = 0$ and $|\Pr[G_4] - \Pr[G_3]| \leq \text{Adv}_{n,q,\sigma_e}^{\text{MLWE}}(\mathcal{B}_1)$.

Summing the hops gives the bound of Theorem 4. The measure-and-reprogram details of the QROM extractor are standard and deferred. $\square$

A.4 Context isolation (Theorem 5)

Openings under $\text{ctx}' \neq \text{ctx}$ contain the masks $z_i(\text{ctx}', \cdot)$ and core terms for ctx' . Applying Lemma 2 at ctx' , these are simulatable from public data and the corrupt shares. The honest masks $z_i(\text{ctx}', \cdot)$ are, by pseudorandomness, independent of the masks $z_i(\text{ctx}, \cdot)$ at the input ctx that no honest party evaluated; and any t -set mixing openings from ctx and ctx' leaves an uncanceled pseudorandom residue (the antisymmetric per-pair terms cancel only when all shares share one ctx). Hence no collection of openings under ctx' reduces the effective threshold under ctx , so the per-context clause of Definition 2 is the only restriction. $\square$

A.5 Rogue ciphertexts and robustness (Theorem 6)

Verify rejects a malformed ciphertext unless the adversary forges an accepting NIZK for a false statement, which reduces to MSIS via soundness. For a well-formed adversarial $\text{ct}^\dagger \in B$, Decap derives each key from that ciphertext's own $\mathbf{u}_j$ and the common Δ ; Δ exposes only the combined value, and cross-leakage to honest keys would require distinguishing the honest $\mathbf{u}_j$ -dependent terms, i.e. breaking MLWE or the PRF. Identifiable abort: each p_i carries π_i ; an invalid share fails π_i (soundness, MSIS) and Open outputs its index, while any t valid shares reconstruct the correct Δ by Lemma 1. $\square$

A.6 Silent post-quantum threshold IND-CCA (Section 8.6, `silentpq/`)

The silent variant replaces the DKG-based core with the Champion–Wu silent scheme (distributed broadcast / monotone-policy encryption) used as a KEM, the object built and validated in `silentpq/`. We record its security at the same level as the core proofs above; the composition and the noise budget are machine-checked in `proofs/silent_qrom.v` (and `docs/silent-pq.md §7`).

Scheme. $\Pi = \text{FO}[\Pi_{\text{cpa}}, G]$. A keyper i silently posts $\mathbf{t}_i = \mathbf{B}\mathbf{y}_i$ (its own $\mathbf{y}_i \in \{0, 1\}^w$; no dealer). To encapsulate, the sender secret-shares the dual-Regev target $\mathbf{p}$ with the small-coefficient $\{0, 1\}$ -LSSS, $\mathbf{p} = \sum_B \mathbf{r}_B$, commits the per-slot values $\mathbf{t}_i + \hat{\mathbf{p}}_i$ through the ww25 matrix commitment $\{\mathbf{C}_j\}$, and outputs the *constant-size* $\text{ct} = (\mathbf{s}^\top \mathbf{B}, \mathbf{s}^\top \sum_j \mathbf{C}_j, \mathbf{s}^\top \mathbf{p} + \mu \lfloor q/2 \rfloor)$, with $K = G(\mu, \text{ad})$. Setup is *transparent* (Section 8.6: public-coin $(\hat{\mathbf{B}}, \mathbf{W}, \mathbf{R})$, Wee–Wu Eq. (4.1), or the public-coin discard ceremony of `wee_commitment.py`). `IsValid` checks a posted key is well formed ($\mathbf{t}_i$ in the image of $\mathbf{B}$); the adversary's corrupt keys must pass it.

Threat model. Threshold IND-CCA with *static* corruption: $\mathcal{A}$ fixes a corrupt set $\mathcal{C}$, $|\mathcal{C}| \leq t - 1$, and the honest indices before seeing keys (*semi-selective*: it also commits the challenge policy/set), then receives the honest public keys, a partial-decryption oracle $\mathcal{O}_{\text{PartDec}}$ on ciphertexts of its choice, and a challenge (ct^*, K_b^*) with K_0^* real and K_1^* uniform. Admissibility forbids a full authorized set of $\mathcal{O}_{\text{PartDec}}$ queries on ct^* .

Theorem 7 (Silent threshold IND-CCA, QROM). *For every QROM adversary $\mathcal{A}$ against Π making q_G queries to G ,*

$$\text{Adv}_{\Pi}^{\text{tcca}}(\mathcal{A}) \leq \epsilon_{\text{sim}} + \epsilon_{\text{fo}} + \text{Adv}_{\Pi_{\text{cpa}}}^{\text{cpa}}(\mathcal{B}),$$

where $\epsilon_{\text{fo}} = O(q_G \delta + q_G / \sqrt{|\mathcal{M}|})$, δ is the decapsulation-error probability, and $\text{Adv}_{\Pi_{\text{cpa}}}^{\text{cpa}}$ is the IND-CPA advantage of the Champion–Wu scheme, which reduces to decomposed LWE (monotone-policy) / ℓ -succinct LWE (broadcast).

Structured hybrid; composition and budget in `silent_qrom.v`. Write $p_i = \Pr[\mathcal{A} \text{ wins } G_i]$.

- $G_0 \rightarrow G_1$ (*share simulation*). Answer each honest $\mathcal{O}_{\text{PartDec}}(i, \text{ct})$ by a simulator that recovers the share $\mathbf{s}^\top \hat{\mathbf{p}}_i$ from the $\leq t-1$ corrupt keys and the public $\{0, 1\}$ -LSSS reconstruction, smudging the missing contribution. By admissibility no context reaches a full authorized set on ct^* , so the simulator never exceeds the threshold; the openings are statistically close, giving $|p_0 - p_1| \leq \epsilon_{\text{sim}}$. (Threshold analogue of Lemma 2.)
- $G_1 \rightarrow G_2$ (*FO in the QROM*). Answer Decap by the Fujisaki–Okamoto re-encryption check with G reprogrammed, eliminating the secret key from decryption (Hofheinz–Hövelmanns–Kiltz 2017; Don–Fehr–Majenz–Schaffner 2019). $|p_1 - p_2| \leq \epsilon_{\text{fo}}$.
- $G_2 \rightarrow G_3$ (*CPA*). Program $\mathbf{p} = \mathbf{B}\mathbf{k}_p$ and the commitments in their simulated form $\mathbf{C}_j = \mathbf{B}\mathbf{K}_j$ (indistinguishable from the real commitments by binding of the ww25 commitment); then $\text{ct}^* = (\mathbf{s}^\top \mathbf{B}, \mathbf{s}^\top \mathbf{B}\mathbf{K}, \mathbf{s}^\top \mathbf{B}\mathbf{k}_p + \mu \lfloor q/2 \rfloor)$ is a function of $\mathbf{s}^\top \mathbf{B}$ alone, so a distinguisher becomes an LWE distinguisher w.r.t. $\mathbf{B}$; $\mathbf{K}^*$ is uniform and $p_3 = \frac{1}{2}$. $|p_2 - p_3| \leq \text{Adv}_{\Pi_{\text{cpa}}}^{\text{cpa}}$, which Champion–Wu reduce to decomposed / ℓ -succinct LWE.

Summing gives Theorem 7. The telescoping $|p_0 - \frac{1}{2}| \leq \epsilon_{\text{sim}} + \epsilon_{\text{fo}} + \text{Adv}_{\Pi_{\text{cpa}}}^{\text{cpa}}$ and the decapsulation budget — correct decoding, hence the bound on δ , closes iff $q > 4z\sqrt{U} \sigma_{\text{share}}$ with $U = \binom{n_{\text{kp}}}{t-1}$ the number of designated-holder shares — are machine-checked in `proofs/silent_qrom.v` (`silent_tcca_advantage`, `qmin_silent`, `sigma_silent_is_sqrtU`, `qmin_grows_with_unqual`; no `Admitted`, no axioms). The three deep inputs — $\text{Adv}_{\Pi_{\text{cpa}}}^{\text{cpa}}$ (the Champion–Wu reductions), ϵ_{fo} (the QROM-FO extractor), and ϵ_{sim} (the smudging distance) — are cited, exactly as Theorem 4 cites its QROM extractor. Security is *selective / semi-selective* and *static*; adaptive corruption is open, as for all lattice ABE/BE. $\square$ $\square$

B Parameter derivation

Table 4 was produced as follows; the arithmetic is elementary and reproducible.

- **Core noise.** For $w - \mathbf{s}^\top \mathbf{u} = e_2 + \mathbf{r}^\top \mathbf{e} - \mathbf{s}^\top \mathbf{e}_1$ over a module of LWE dimension $n = Nk$, each inner product is a sum of n products of width- σ_e terms, so $\sigma_{\text{dec}} \approx \sqrt{2n} \sigma_e^2$. At $n = 1536$, $\sigma_e = 3.2$ this is ≈ 568 .
- **Flooding.** The Rényi requirement of Lemma 2 with order $a = 2$ and query bound $Q = 2^{12}$ gives $\sigma_{\Pi} = \sigma_{\text{dec}} \sqrt{2\pi a Q N} \approx 2.1 \cdot 10^6$ (the $\sqrt{N}$ is the ring correction of Remark 5; without it the masked openings would not be simulatable).
- **Combination.** With the replicated/CNF $\{0, 1\}$ -sharing the recovery coefficients are in $\{0, 1\}$, so $\kappa = \sqrt{t}$ for every t -subset; with the exact zero-sharing ($\beta_F = 0$, Remark 4) the combined width is $\sigma_{\text{tot}} = \sqrt{\sigma_{\text{dec}}^2 + \kappa^2 \sigma_{\Pi}^2}$. At the deployable committee $(t, n) = (7, 13)$ this is $\kappa = \sqrt{7} \approx 2.65$ and $\sigma_{\text{tot}} \approx 5.5 \cdot 10^6$; on the ring $N=256, k=6, q=2^{31}-511$ ($n=1536$, ≈ 157 -bit core-SVP) the budget closes with vast margin and every random 7-of-13 committee decapsulates the whole batch end to end (`tacet.params.DEPLOY`; `bench/deploy_validation.py`; locked in by `tests/test_tacet.py`). The honest cost is per-party key material of $\binom{n-1}{t-1} = 924$ ring tuples at $(7, 13)$.

Why not a larger committee (the standing open problem). One would like a large committee such as $(100, 67)$. That needs a sharing whose recovery norm κ stays small at $t = 67$ while the shares stay short. No cited construction is known to deliver both: implementing the Pilvi sharing on this ring and measuring κ (`bench/pilvi_sharing.py`, reconstruction residual $< 10^{-12}$) shows it is *exponential* in t — $\kappa \approx 6 \cdot 10^{20}$ already at $t=16$, $\approx 21.5 \times$ per unit t , $\sim 10^{67}$ near $t \approx K/2$ — matching Pilvi’s own paper (concrete instantiations stop at $K=32$, $q \sim 2^{84}$). So the $\kappa = 16.4$ one would want at $(67, 100)$ is *not* delivered by any known low-norm sharing (it is assumed, not achieved), and $q_{\text{min}} \propto \kappa$ would force $q \gg 2^{31}$. The replicated sharing keeps $\kappa = \sqrt{t}$ but its per-party material is $\binom{n-1}{t-1}$, which is 924 at $(7, 13)$ and blows up at $(100, 67)$. A sharing that is *simultaneously* low-norm and small-share at large t is the open problem; the construction itself is unaffected. See `AUDIT.md` §5.

- **Modulus and failure.** For overall failure $\leq 2^{-128}$ over $N = 256$ coefficients we need $\text{erfc}(z/\sqrt{2}) \leq 2^{-(128 + \log_2 N)}$, i.e. $z \approx 13.5$, so $q > 4z \sigma_{\text{tot}}$ suffices. At $\sigma_{\text{tot}} \approx 5.5 \cdot 10^6$ the NTT prime $q = 2^{31} - 511$ gives realised thresh-

old $z_{\text{real}} = (q/4)/\sigma_{\text{tot}} \approx 98.5$, far above 13.5 (realised failure $\leq 2^{-6998}$) — a deliberate margin against the approximations here. (A power-of-two $q = 2^{32}$ is not NTT-friendly, Section 10; the code uses the prime.)

- **Sizes.** A ring element is $N \log_2 q$ bits, so ct ($k+1$ elements: $\mathbf{u}$ and w) is ≈ 7 KB, the replicated per-party key material is $\binom{n-1}{t-1}$ ring tuples (924 at $(7, 13)$), and a per-ciphertext opening $\rho_{i,j}$ (one element) is ≈ 1 KB, so a batch opening is $\approx |B|$ KB ($O(|B|)$). NIZK proofs are separate and verify off the critical path.
- **Security target.** The public-key instance is MLWE at $(n=1536, q \approx 2^{32}, \sigma_e=3.2)$. A primal-uSVP estimate (the 2016 estimate: smallest BKZ block β with $\sqrt{\beta} \sigma_e \leq \delta^{2\beta-d-1} q^{m/d}$ over the embedding dimension $d = m + n + 1$, optimised over m , with δ the root-Hermite factor of β) gives $\beta \approx 514$, hence ≈ 150 -bit classical (0.292β) and ≈ 136 -bit quantum (0.265β) core-sieve cost. We corroborate this with a standalone estimator validated against Kyber and FrodoKEM (Section 10): it reproduces $\beta = 514$ at $q = 2^{32}$, gives $\beta = 538$ (142-bit quantum) at the implementation prime $2^{31} - 511$, and finds the dual attack ($\beta \approx 630$) non-binding. Core-SVP is the optimistic model; a full lattice-estimator run with a gate-count model (which raises the figure) remains advisable before deployment.